

From Pixels to Visual Intelligence



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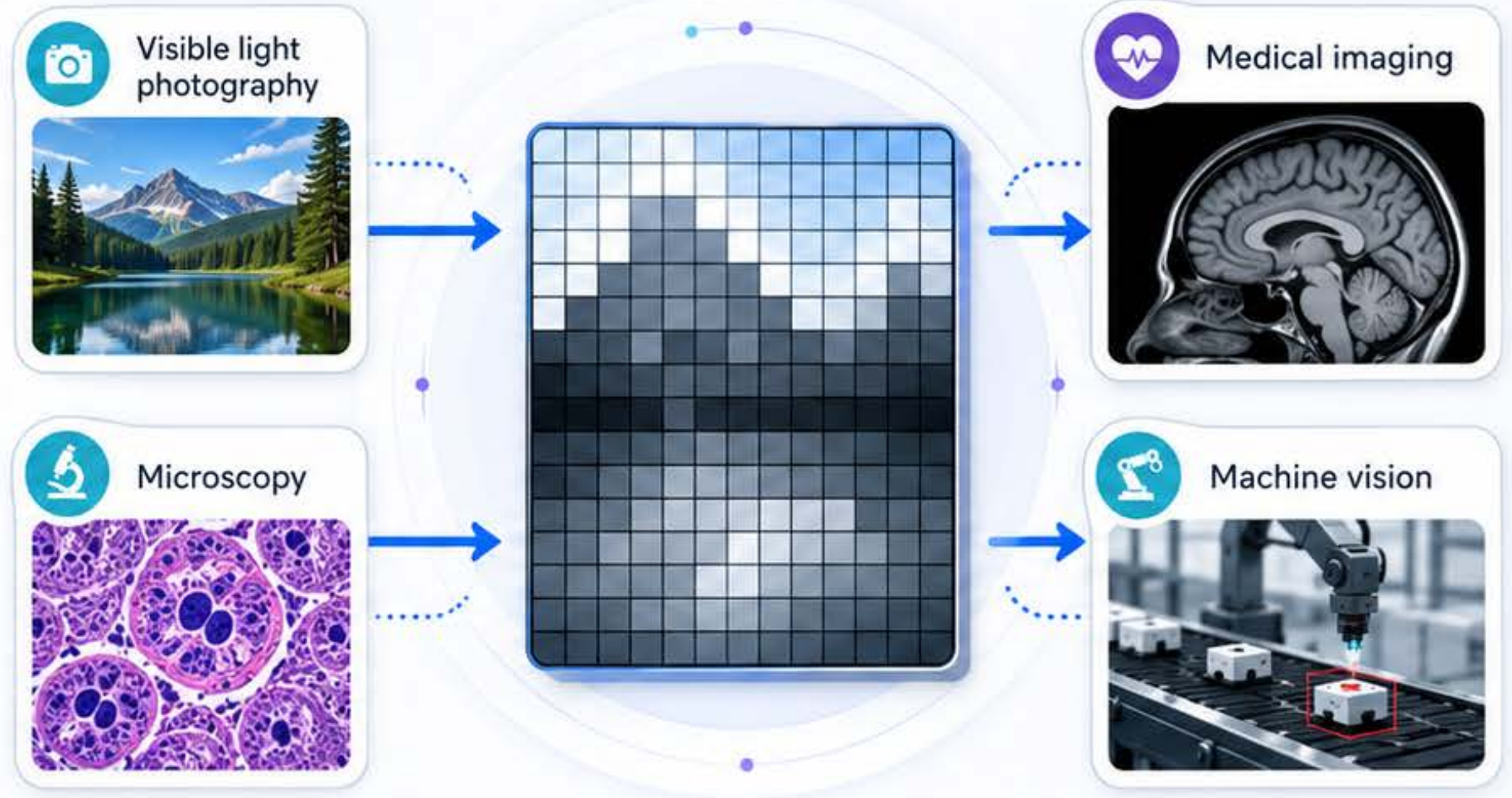


GenAI BootCamp with emphasis
on **Image Processing**



Foundations of Image Processing

How images are represented, digitized, and used for analysis



Represent

Images encode spatial information.



Digitize

Sampling and quantization turn scenes into data.



Analyze

Processing supports enhancement, measurement, and interpretation.



Visible light



Microscopy



Medical imaging



Computer vision

An Image Is Spatially Organized Information



An image records a **measured quantity** across a spatial domain.

- ✓ Values gain meaning from their spatial relationships.
- ✓ Edges, textures, regions, and objects emerge from local patterns.
- ✓ Images support both visualization and quantitative analysis.



Visible image



X-ray / medical image



Fluorescence microscopy



Thermal or depth map



Images may encode brightness, color, X-ray attenuation, fluorescence, depth, temperature, and more.



pixels

Individual samples



neighborhoods

Local context and structure



regions

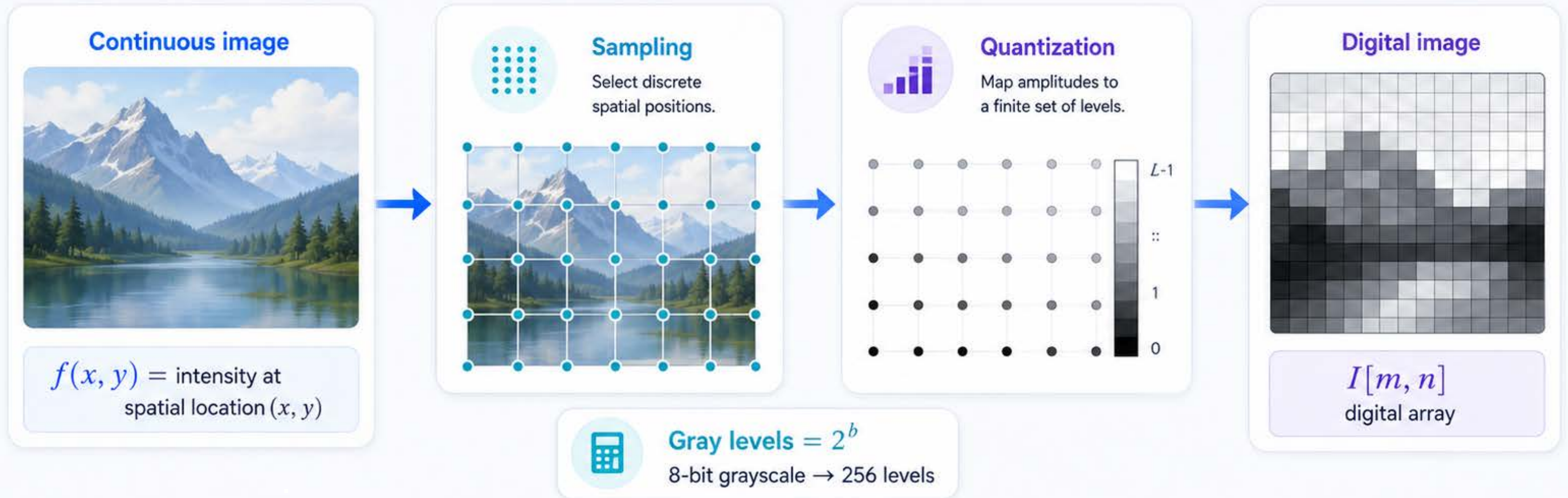
Spatially coherent areas



objects

Meaningful entities

From Continuous Scene to Digital Image



Continuous:
coordinates and
intensity vary smoothly.



Sampling
controls spatial
resolution.



Quantization
controls intensity
precision.

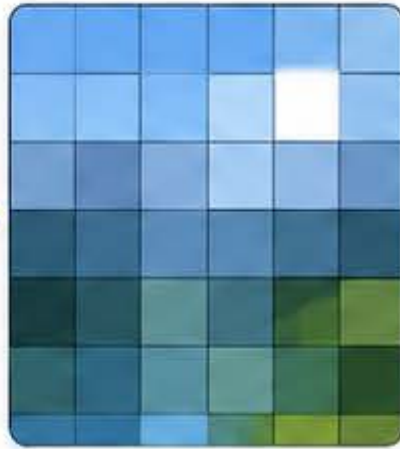
A Visual Example of Sampling and Quantization



Sampling



Continuous scene



Coarse sampling
(low resolution)



Finer sampling
(higher resolution)



Too few samples → lost detail / aliasing



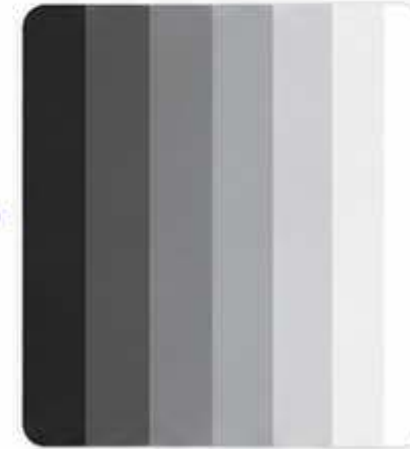
More samples → better spatial detail



Quantization



Continuous intensity



4-level
quantization



8-bit (256-level)
quantization



Few levels → visible banding



More levels → smoother intensity variation



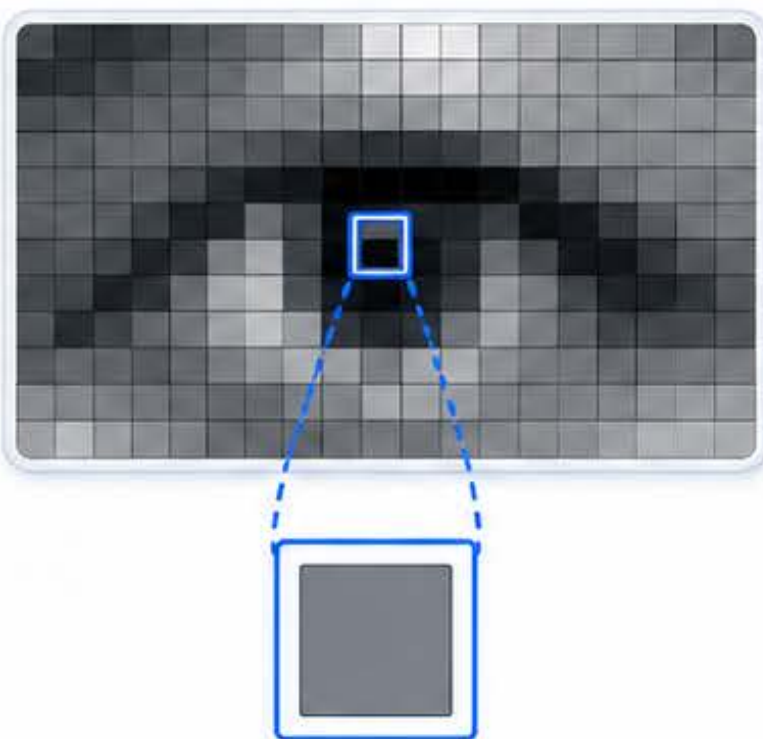
Spatial resolution and **intensity precision** are different design choices.

Pixels, Gray Levels, and Neighborhoods



Pixel

A pixel is the smallest addressable element in a digital image.

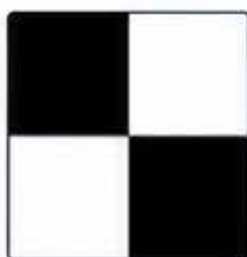


Gray levels and bit depth

Bit depth determines how many gray levels can be represented.



1-bit: 2 levels



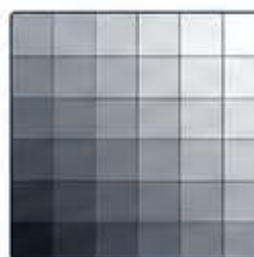
0 or 1

8-bit: 256 levels



0-255

16-bit: higher measurement precision

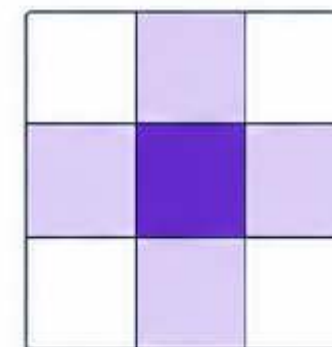


0-65,535



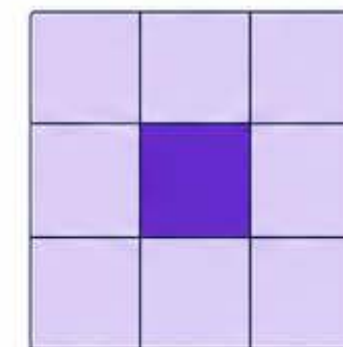
Neighborhoods

4-neighbor



Center pixel

8-neighbor



Neighbor

Local neighborhoods support smoothing, edge detection, and segmentation.



A single pixel rarely has meaning by itself; **context comes from surrounding pixels and regions.**

Where Image Processing Fits



Computer graphics

Creates images from models and rendering rules.



Typical question:

How can an image be generated?

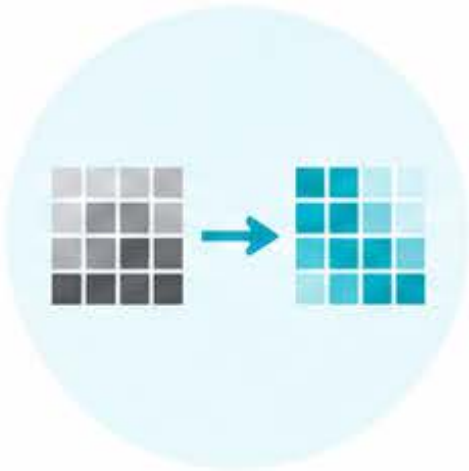


Image processing

Transforms existing images to improve, restore, compress, measure, or prepare them.



Typical question:

How can this image be improved or transformed?



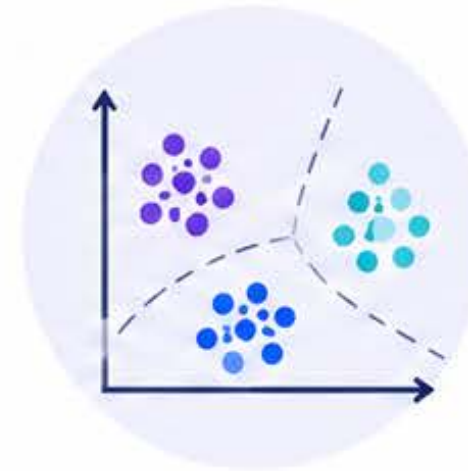
Computer vision

Extracts structure, objects, motion, and semantic information.



Typical question:

What is present, where is it, and what is happening?



Pattern recognition

Assigns classes or decisions from features or image data.



Typical question:

Which category does this pattern belong to?



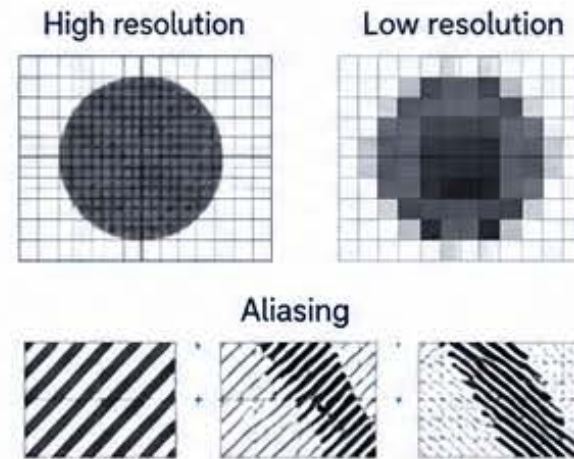
Real applications often combine these fields in one pipeline; image understanding adds domain-specific meaning.

Practical Considerations for Reliable Imaging



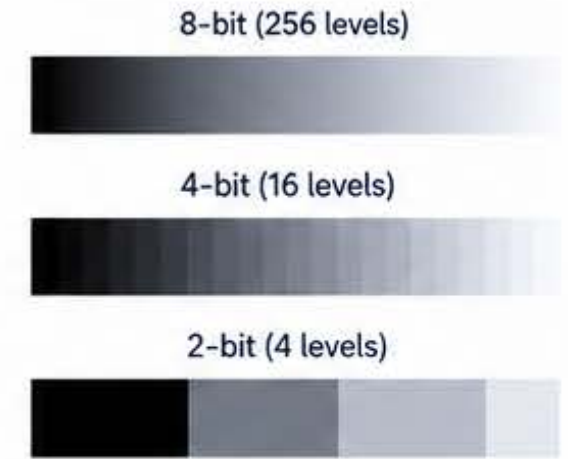
Spatial resolution

If sampling is too coarse, fine detail is lost and aliasing may appear.



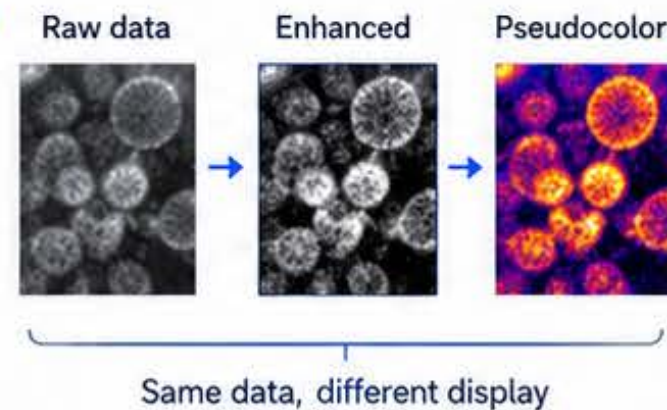
Bit depth

Too few intensity levels reduce precision and can create banding.



Display versus data

Brightness, contrast, or color mapping can help viewing but may mislead measurement if the original data are not preserved.



Metadata and context

Exposure, magnification, channel identity, calibration, and processing history are essential for interpretation.



A visually attractive image is not automatically a scientifically valid result.

Applications Across Imaging Domains



Restoration

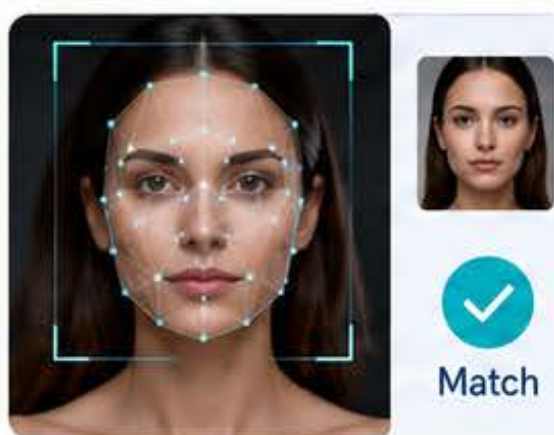


Task: Denoising, deblurring, artifact reduction

Goal: Recover a more useful image.



Biometrics / face analysis

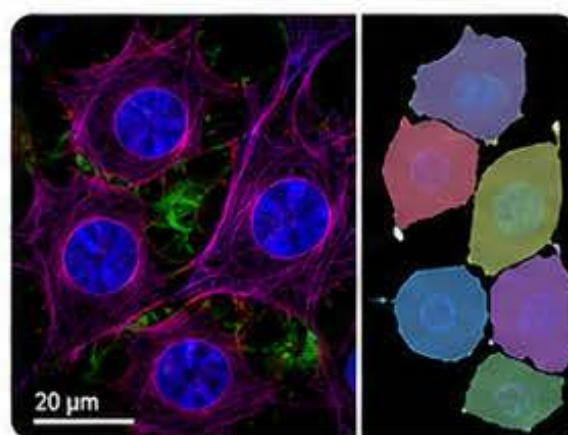


Task: Detection, tracking, matching

Goal: Locate or verify identity.



Microscopy

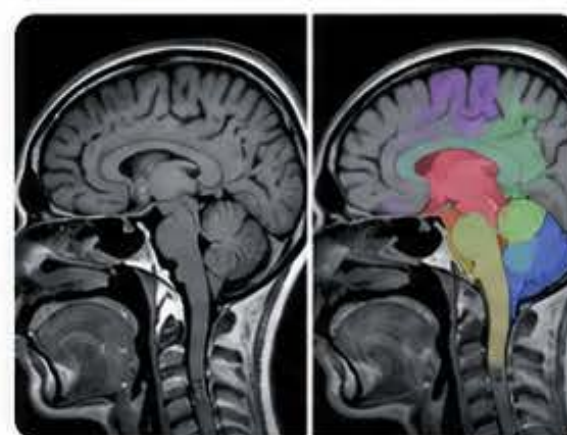


Task: Enhancement, segmentation, measurement

Goal: Quantify structures and processes.



Medical imaging

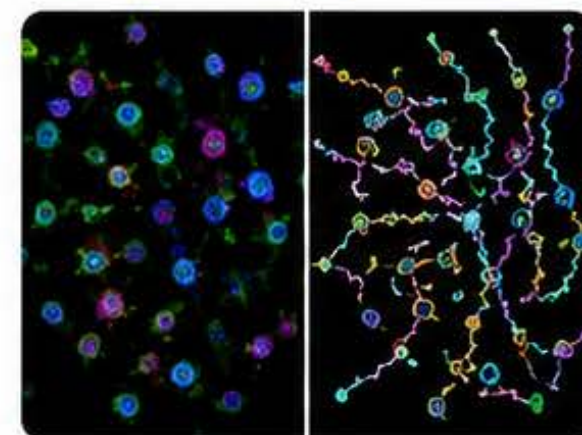


Task: Segmentation, registration, retrieval

Goal: Support diagnosis and planning.



Bioimage analysis



Task: Cell detection, tracking, classification

Goal: Turn biological images into reproducible insights.



Machine learning and deep learning extend classical workflows for **denoising, segmentation, tracking, and classification**.

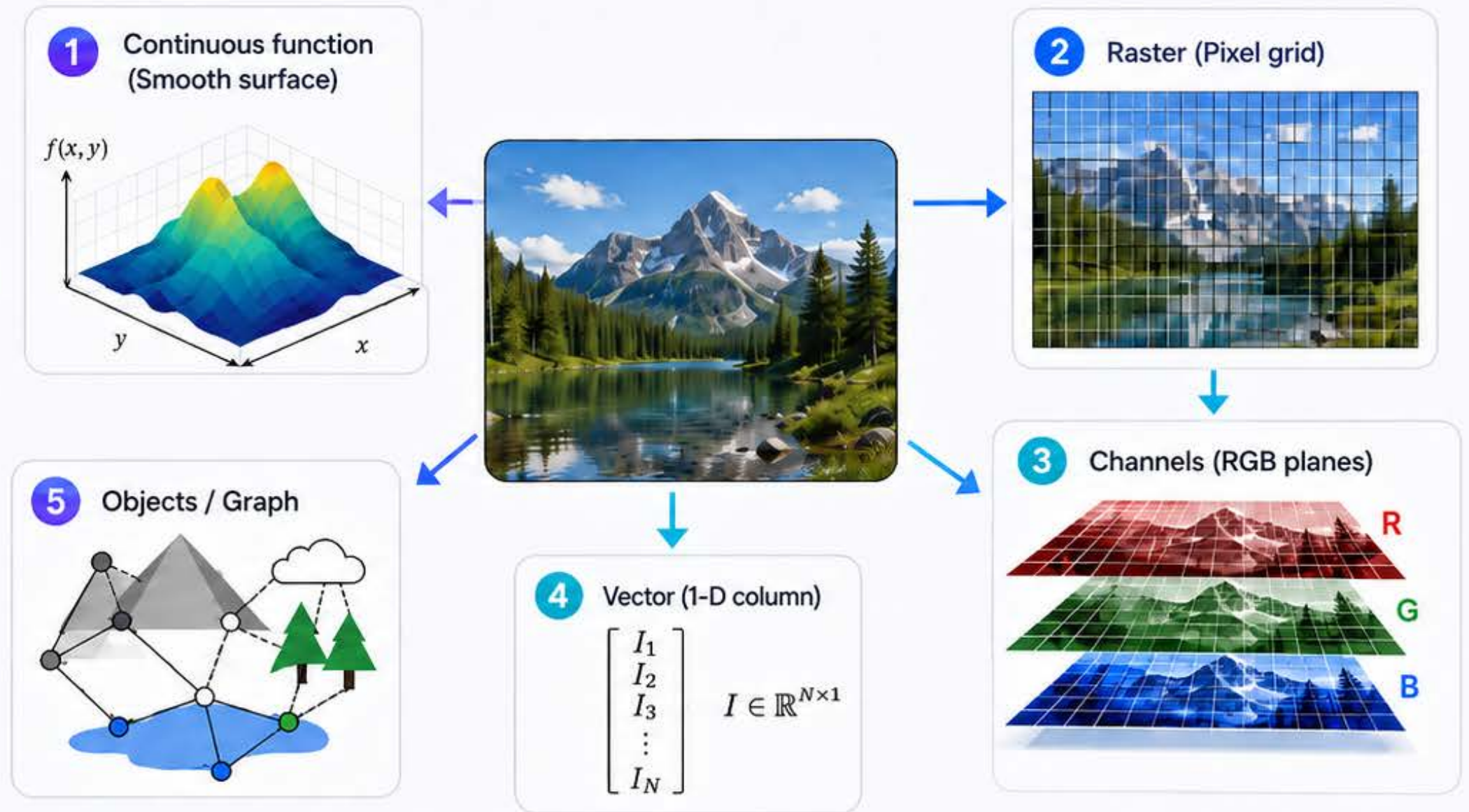
From Acquisition to Interpretation



Front-end hardware influences what can be measured: resolution, focus, field of view, spectral selectivity, noise, and dynamic range.

Image Representation in Image Processing

How the same image can be organized as pixels, channels, vectors, objects, and surfaces



Preserve information

Representation determines what image details are retained.



Enable algorithms

Different representations make filtering, segmentation, and learning easier or harder.



Match the task

Choose the form that best supports measurement, recognition, or interpretation.



Representation is not cosmetic; it shapes what later processing can do.

What Is an Image Representation?



What is stored?

- pixels, channels, objects, features, probabilities.



How are values related?

- spatial neighborhood, geometry, similarity, connectivity, uncertainty.



What does it support?

- filtering, segmentation, measurement, recognition, learning.

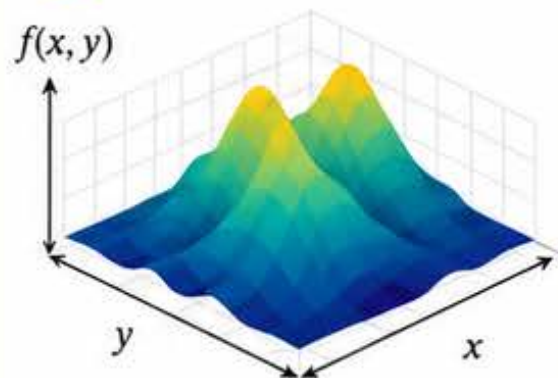


Definition

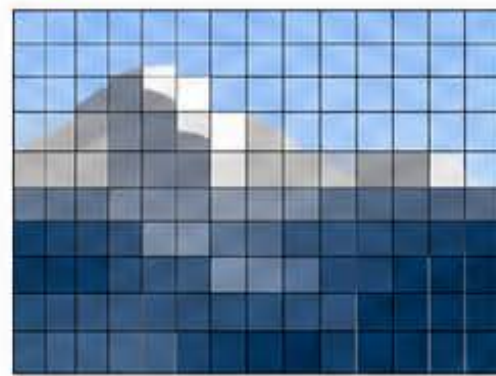
An image representation is a formal way of organizing image information so it can be stored, displayed, processed, measured, or interpreted.



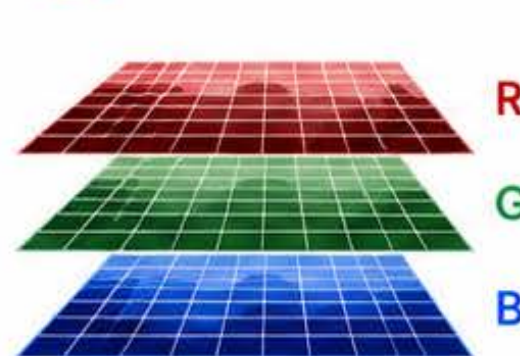
Continuous function



Raster grid



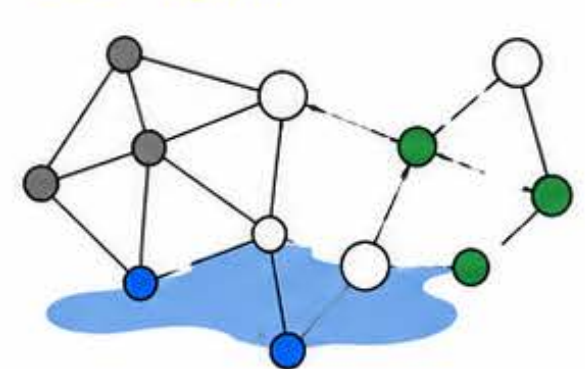
Channel stack



Vector / object



Graph / probabilistic model

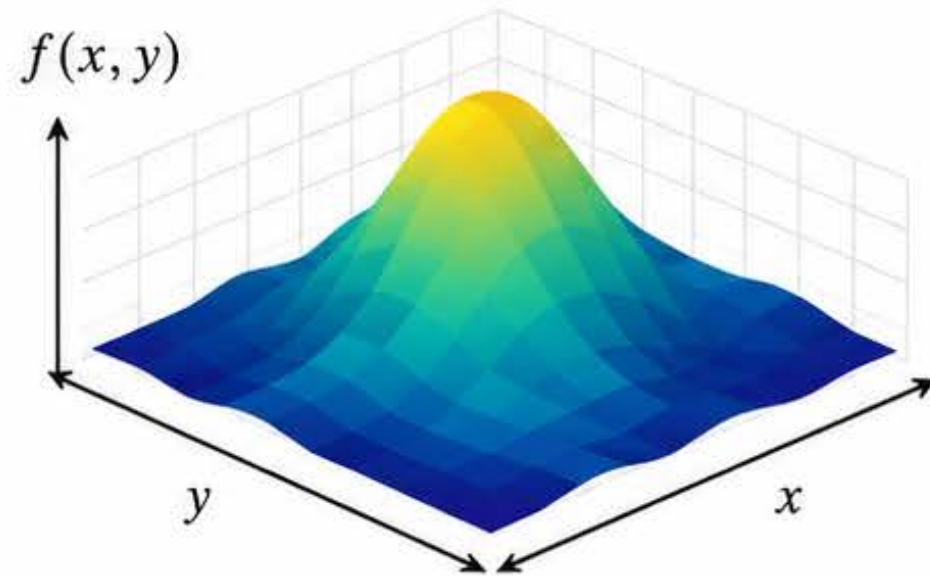


Representation is a computational choice that makes some operations natural and others difficult.

From Continuous Coordinates to Digital Indices

Continuous models describe intensity over space; digital images store sampled values on a grid.

1 Continuous image representation



- Coordinates refer to spatial position.
- Intensity varies continuously.
- Useful for optics, interpolation, gradients, and mathematical analysis.

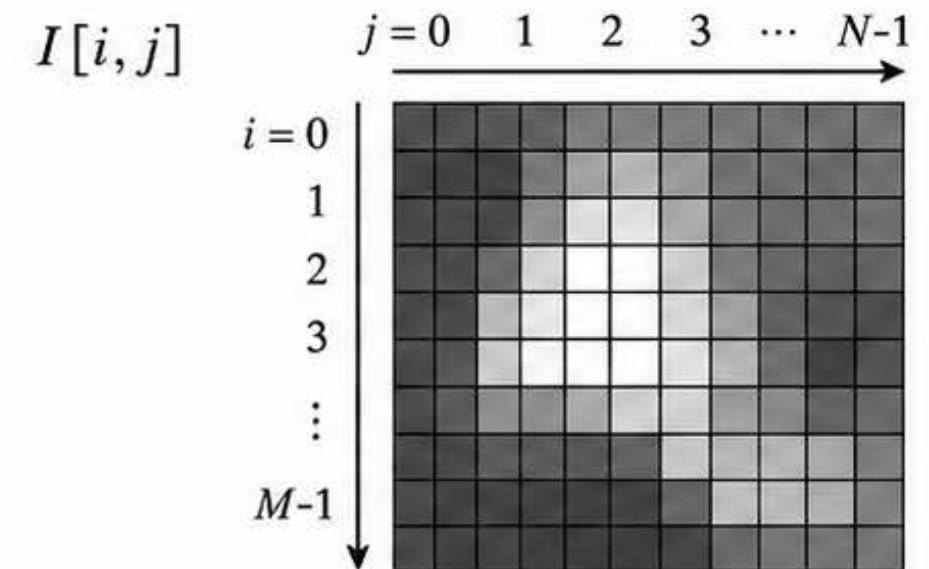


sampling



quantization

2 Digital image representation



- Indices are integers.
- Each pixel stores one or more values.
- Suitable for storage, filtering, segmentation, and computation.



Indexing pitfall: row i increases downward, column j increases to the right; this differs from the usual $x - y$ plot.



Note: In microscopy, x and y may represent physical distances such as micrometers before digitization.

A Small Image as Data

1 Raster (Pixel grid)

		Column (x)			
		1	2	3	4
Row (y)	1	18	42	75	92
	2	30	96	180	210
	3	22	64	120	160
	4	10	40	88	140

$I[2,3]$

Raster view

2 Matrix (Intensity values)

$$I \in \mathbb{R}^{4 \times 4}$$
$$\begin{bmatrix} 18 & 42 & 75 & 92 \\ 30 & 96 & 180 & 210 \\ 22 & 64 & 120 & 160 \\ 10 & 40 & 88 & 140 \end{bmatrix}$$

Matrix view

3 Vectorization (Row-major order)

$$x \in \mathbb{R}^{16 \times 1}$$

$\begin{bmatrix} 18 \\ 42 \\ 75 \\ 92 \\ 30 \\ 96 \\ 180 \\ 210 \\ 22 \\ 64 \\ 120 \\ 160 \\ 10 \\ 40 \\ 88 \\ 140 \end{bmatrix}$

Vectorization
rearranges the
matrix into a
1-D feature vector.

Vector view

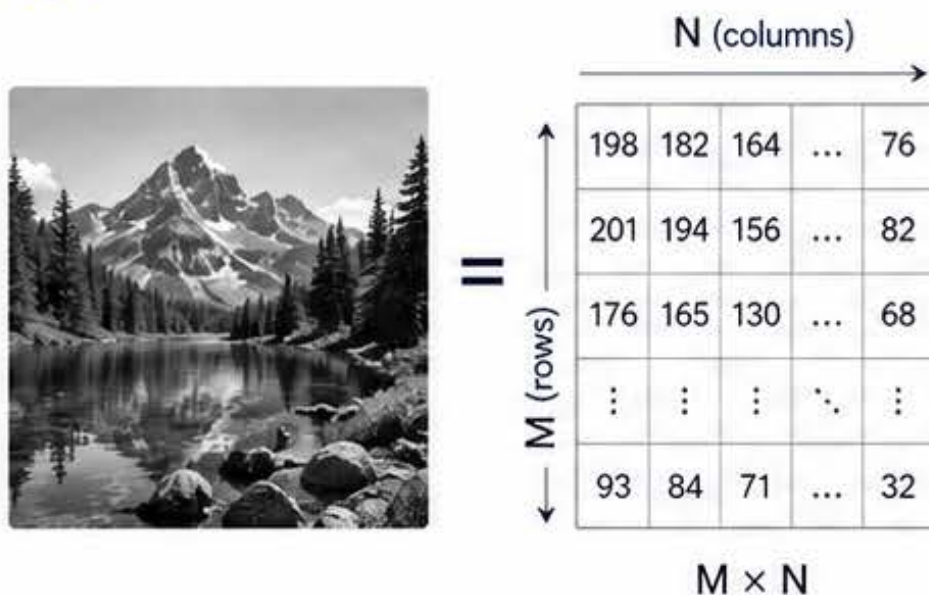


Vectorization is useful for linear algebra and machine learning, but the original image shape must be known to recover spatial structure.

Raster, Matrix, and Multi-Channel Arrays

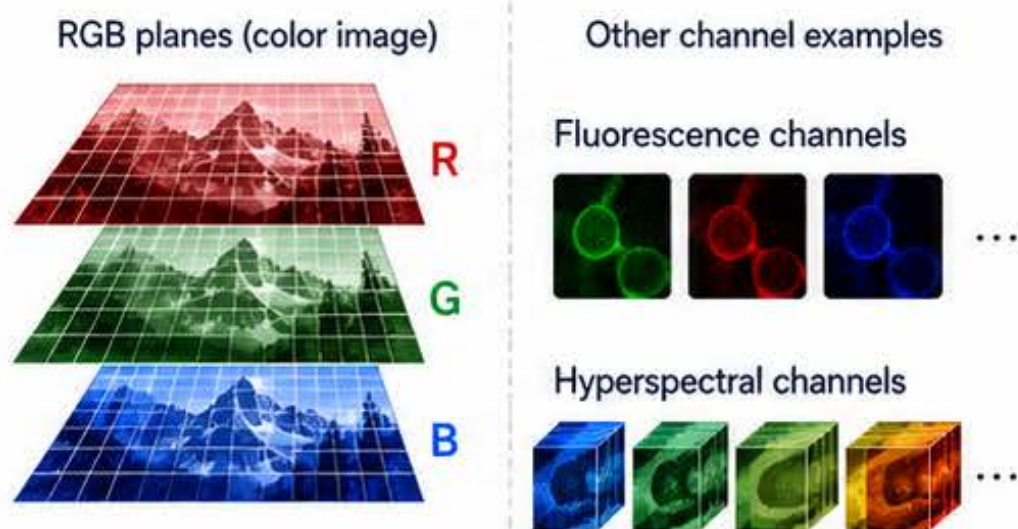
The first major representation family: images as sampled grids of values.

1 Grayscale matrix



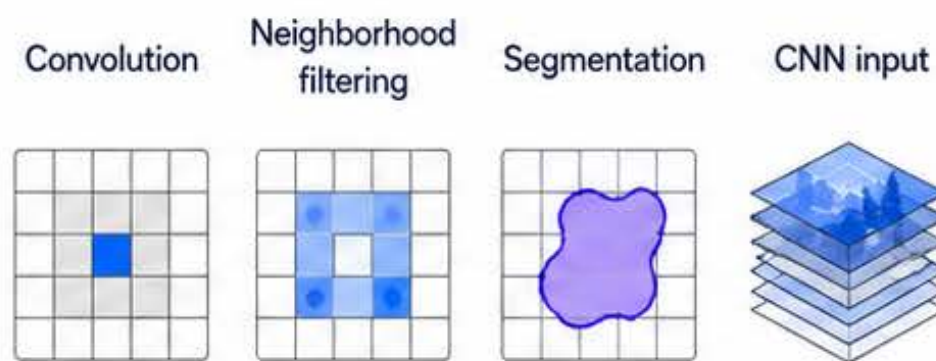
- One intensity value per pixel.
- Natural form for filtering and measurement.

2 Color and channel stacks



- A color or scientific image stores multiple values per pixel.
- Often organized as height $\times$ width $\times$ channels.

3 Why raster dominates



- Local neighborhoods are explicit.
- Ideal for sensed images such as photographs, microscopy, and medical scans.
- Limitation: data size grows with resolution and channels.



Strengths



Local operations



Direct sampling



Standard input to algorithms



Limits



Resolution-dependent



Can be large

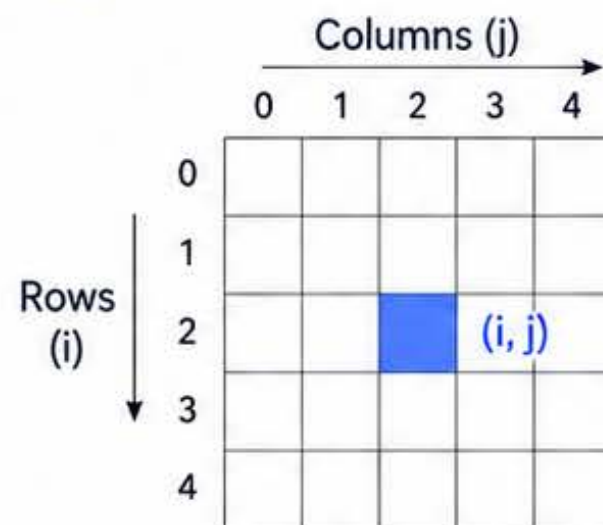


Enlargement reveals pixels

Practical Representation Issues

Limitations, errors, and interpretation concerns that affect reliable use of image representations.

1 Indexing conventions



Confusing x-y geometry with i-j array indexing causes common programming errors.

2 Bit depth and dynamic range

8-bit (256 levels)

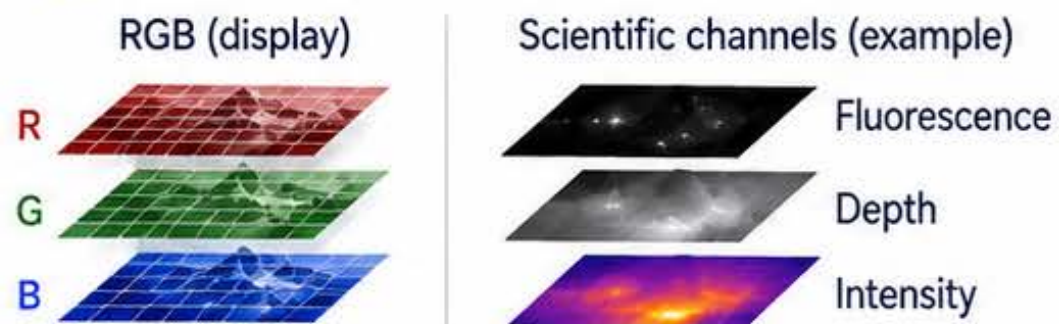


16-bit (65,536 levels)



Quantitative images may need more precision than ordinary display.

3 Channel identity and calibration



A channel is a measured signal, not always a display color.

4 Metadata and context



Preserve calibration, channel labels, and processing history for reliable measurement.

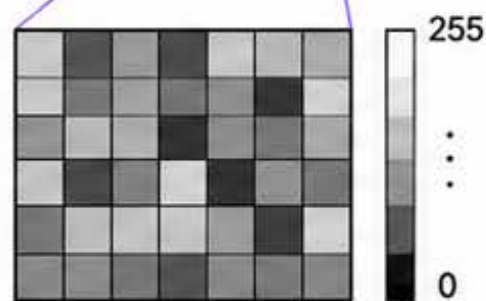
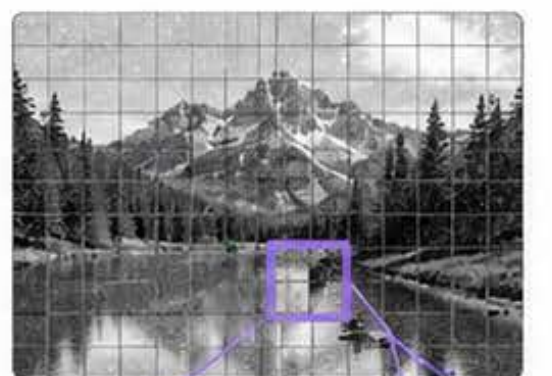


Representation can introduce bias if important information such as scale, connectivity, noise characteristics, or channel meaning is discarded.

Representations in Real Applications

1 Photography / restoration

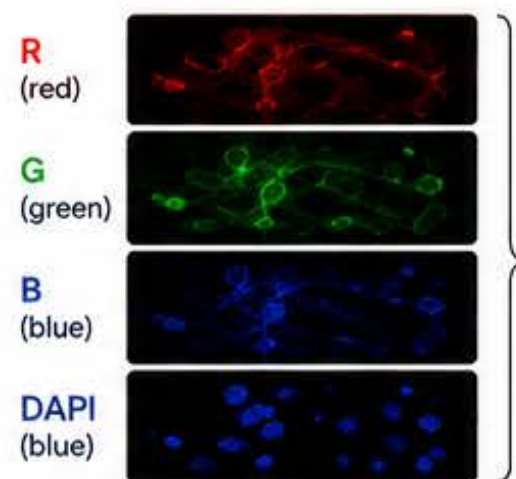
Raster pixels or grayscale arrays used for denoising and enhancement.



Denoise → Enhance

2 Microscopy

Channel stacks preserve biological signals for measurement.

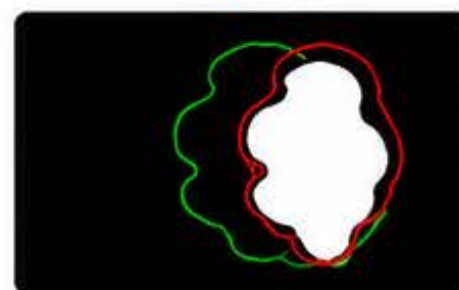
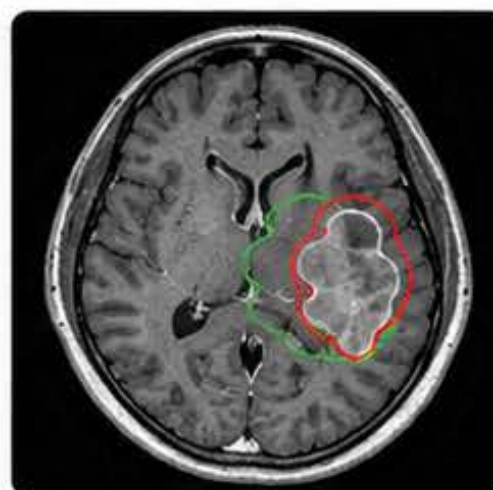


Segmentation mask



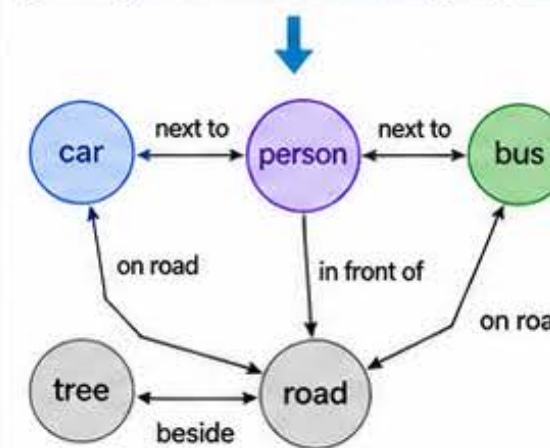
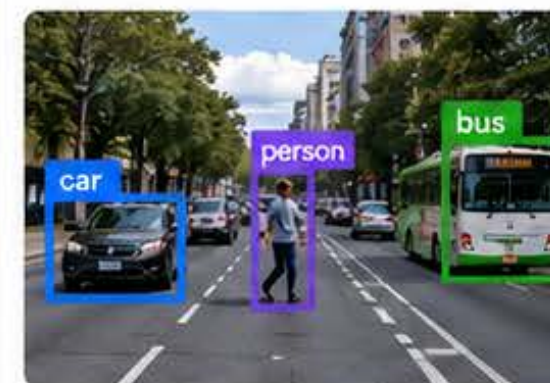
3 Medical imaging

Representation supports segmentation, registration, and analysis.



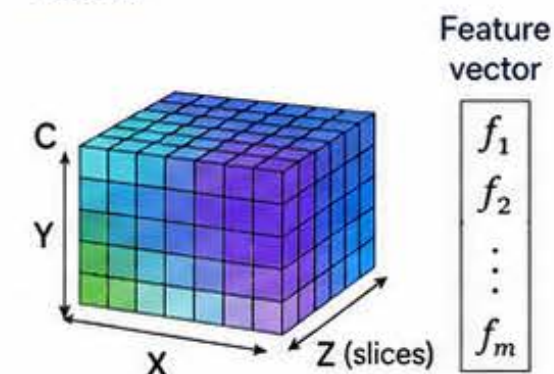
4 Computer vision

Object and graph views support detection and relationships.



5 Bioimage analysis / learning

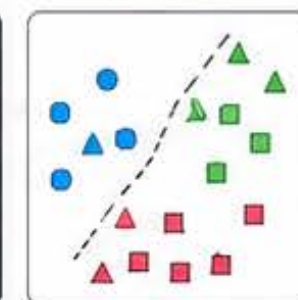
Learning models use tensors, features, and labels.



Cell tracking



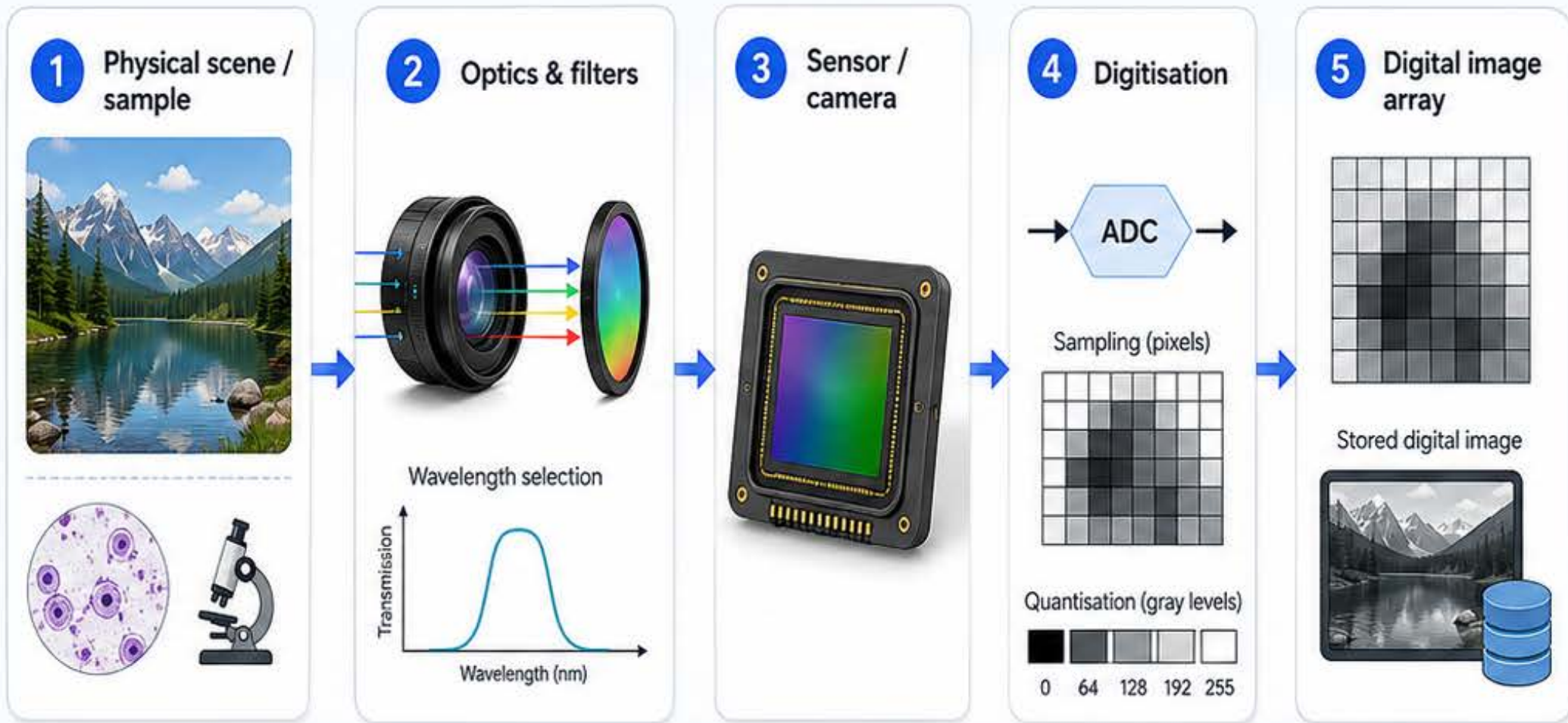
Classification



Applications often move through multiple representations rather than relying on only one form.

Image Acquisition and Digitisation

From physical scenes and biological samples to reliable digital image data.



Capture the right signal

The modality and optics determine what information reaches the detector.



Control quality early

Resolution, noise, contrast, and dynamic range are shaped during acquisition.



Support later processing

Reliable acquisition improves enhancement, segmentation, measurement, and learning.



Good acquisition sets the ceiling for all later image processing.

What Is Image Acquisition?

Defining the process and the roles that turn physical information into image data.



Image acquisition is the process of sensing physical information from a scene, object, or sample and converting it into image data that can be stored, displayed, transmitted, or processed.



Optics

Form or select the signal that reaches the detector.



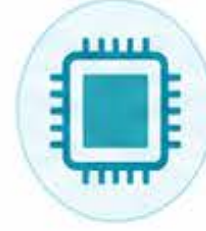
Focus



Field of view



Wavelength selection



Sensor

Convert light, radiation, heat, or depth into an electrical signal.



Acquisition interface

Digitise, transfer, buffer, and organise the signal into an image array.



What is measured?

A physical signal such as reflected light, fluorescence, thermal emission, attenuation, or depth.



Why it matters

Acquisition determines contrast, noise, resolution, and dynamic range.



Downstream impact

Better capture supports filtering, segmentation, measurement, and recognition.

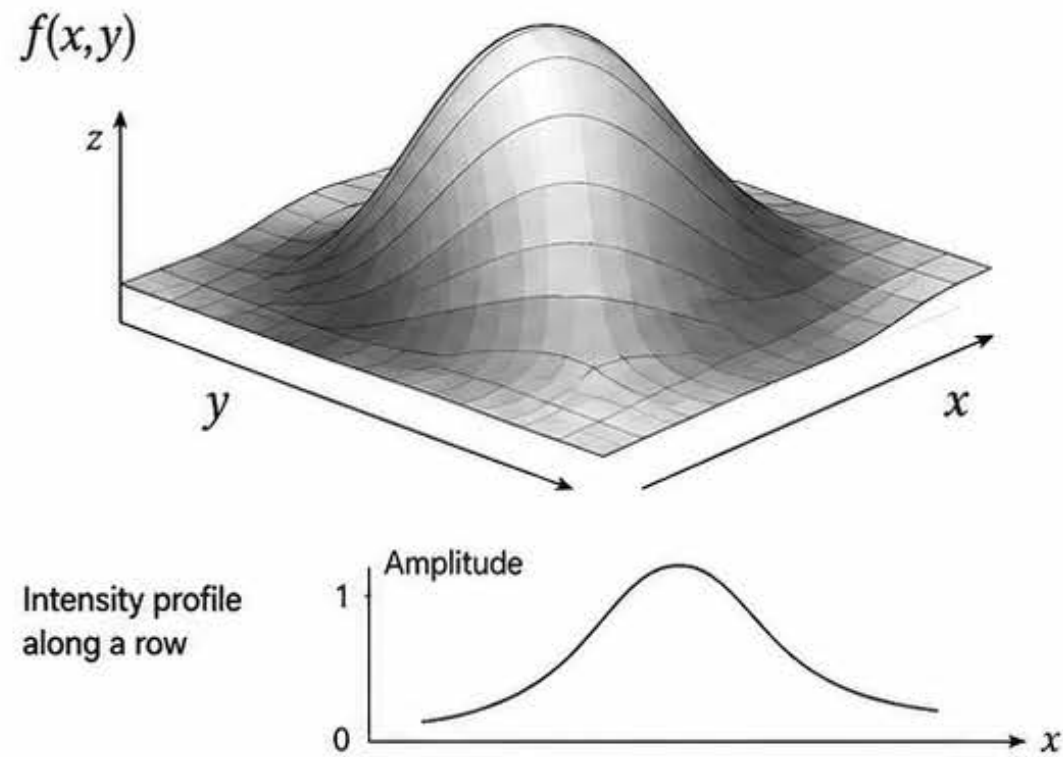


Acquisition is more than taking a picture; it is the first computational step in imaging.

From Continuous Signal to Digital Image

Digitisation requires sampling of spatial position and quantisation of signal amplitude.

Continuous image



- Signal varies continuously over space.
- Useful ideal model before digitisation.

Sampling

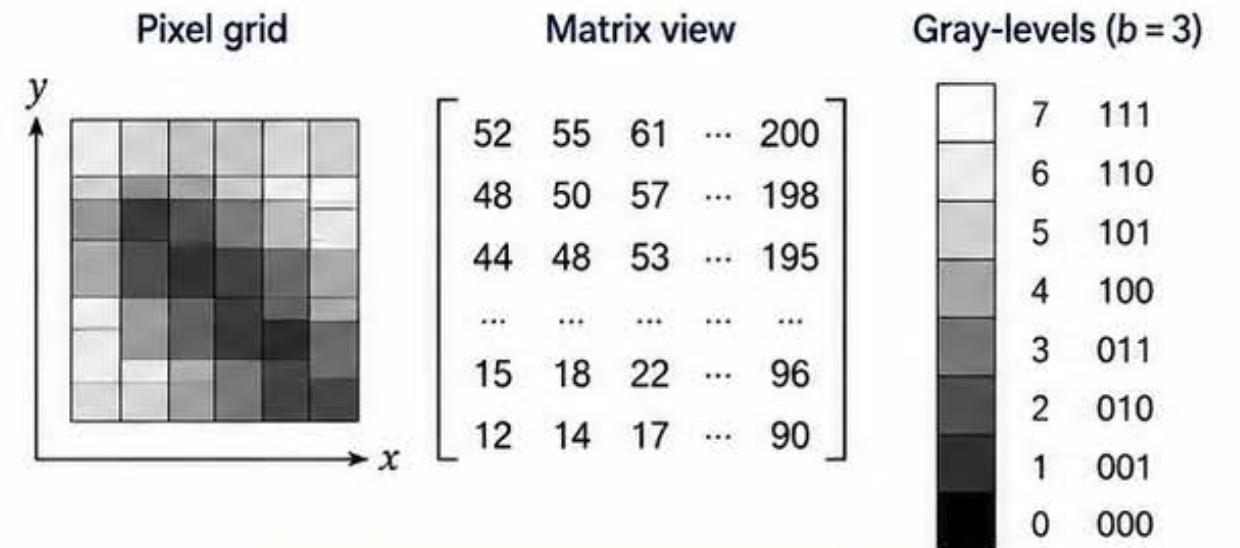
sample interval $\Delta x, \Delta y$



Quantisation

ADC maps values to discrete levels

Digital image



Number of gray levels = 2^b

bit depth b

pixel values

- Sampling answers where to measure.
- Quantisation answers how finely to code intensity.

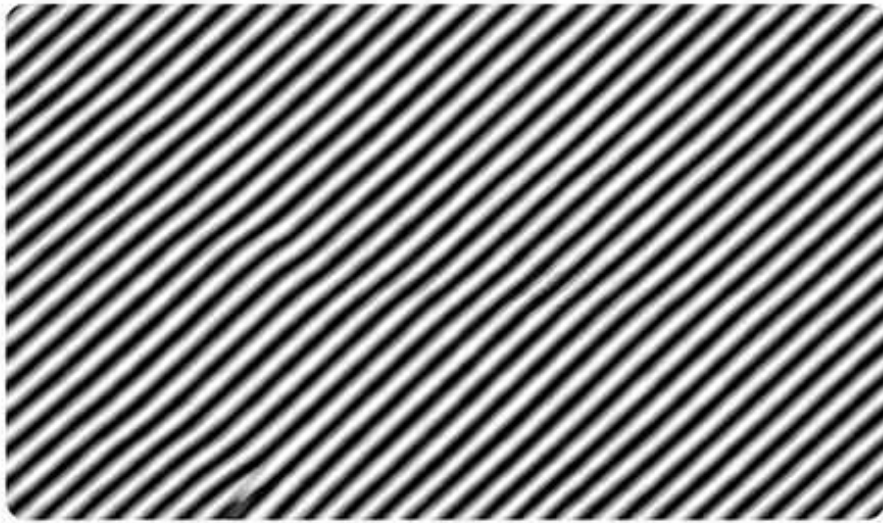


A digital image is formed only after both position and amplitude have been discretised.

Sampling, Tessellation, and Aliasing

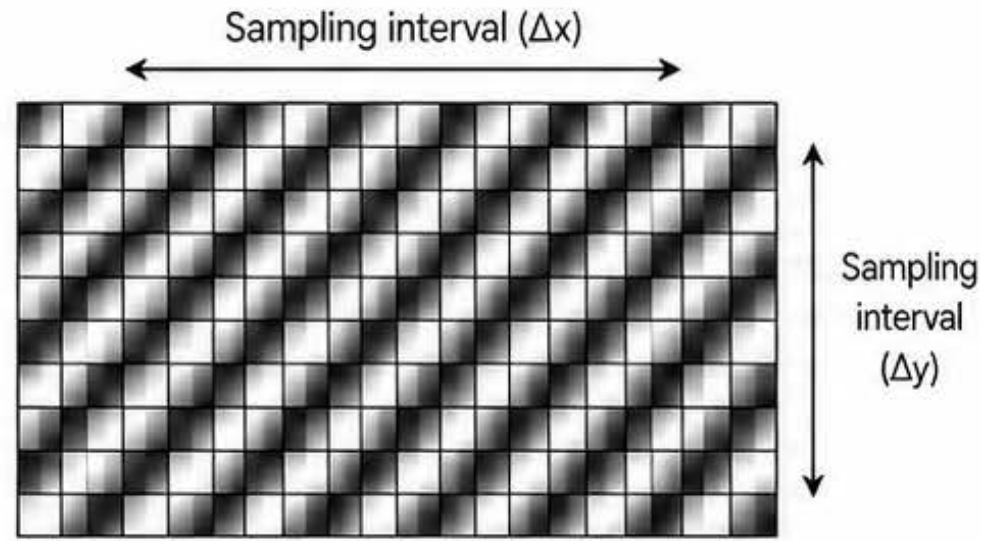
Spatial detail must be sampled densely enough to avoid false structure.

Continuous pattern



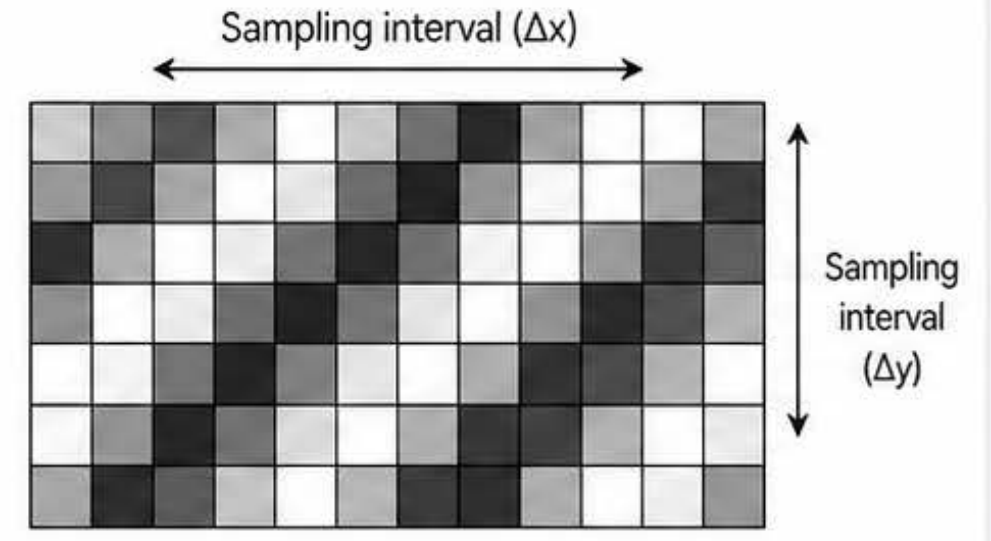
Original fine detail

Adequate sampling

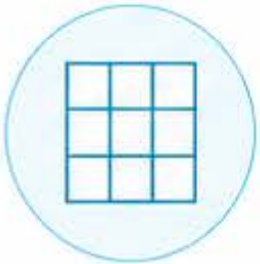


detail represented faithfully

Undersampling

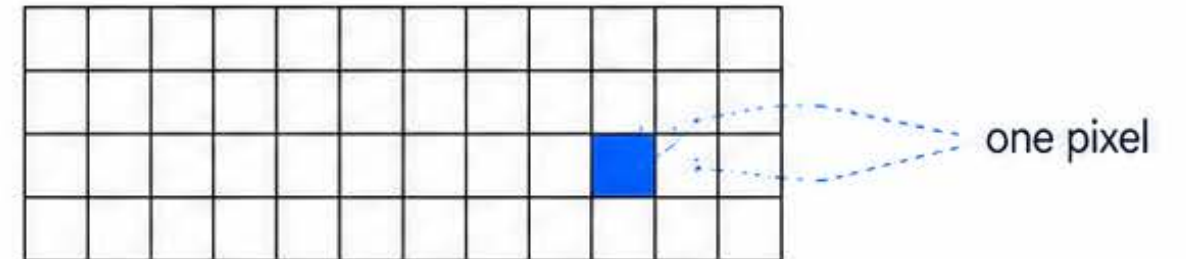


false structure appears



Tessellation

- Tessellation partitions the image plane into pixels.
- Rectangular grids are common because they are simple to sample, store, and process.



Aliasing is not just loss of detail; it can create patterns that were not present in the original scene.

Cameras, Sensors, Lenses, and Filters

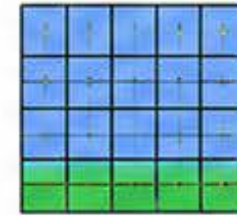
Acquisition hardware determines what the sensor sees and how reliably it is recorded.



Camera

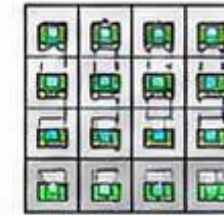
- Controls exposure timing, readout, synchronisation, and transfer.
- Packages the sensor into a usable acquisition system.

CCD



Charge shifted to output

CMOS



Per-pixel amplifiers



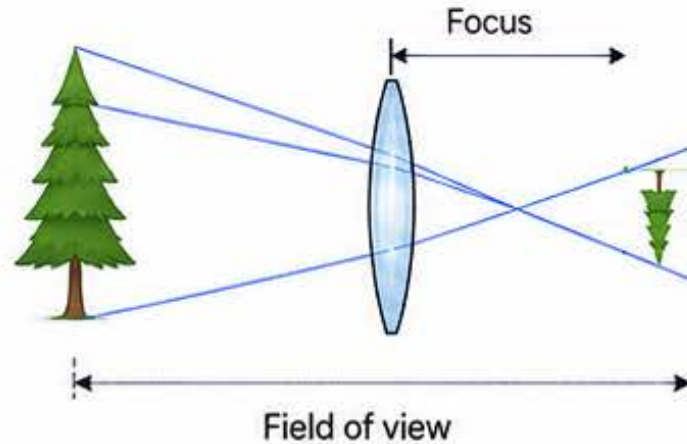
Sensor

- Converts physical energy into an electrical signal.
- Choice depends on sensitivity, speed, noise, and cost.



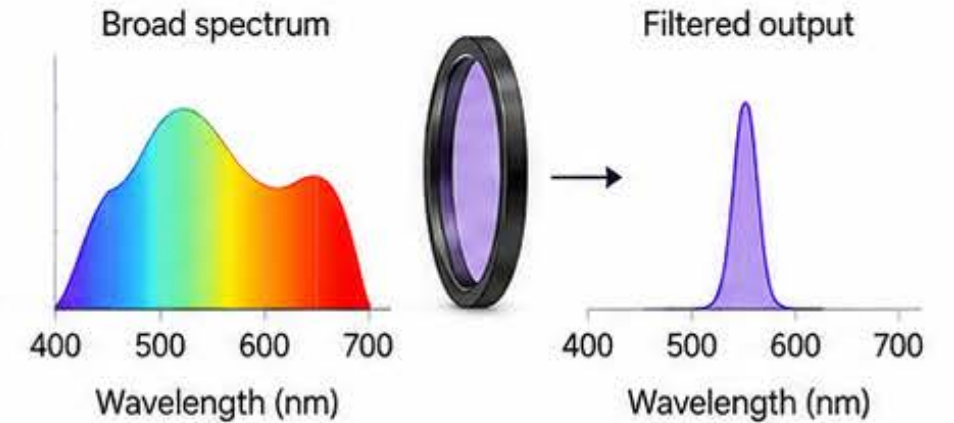
Lens

- Sets focus, magnification, field of view, and distortion.
- Poor image formation cannot be fully repaired later.



Optical filters

- Pass selected wavelengths while blocking others.
- Essential for color imaging, fluorescence, infrared, and ultraviolet work.



Focus

Determines what is sharp.



Magnification

Determines size on the sensor.



Spectral selectivity

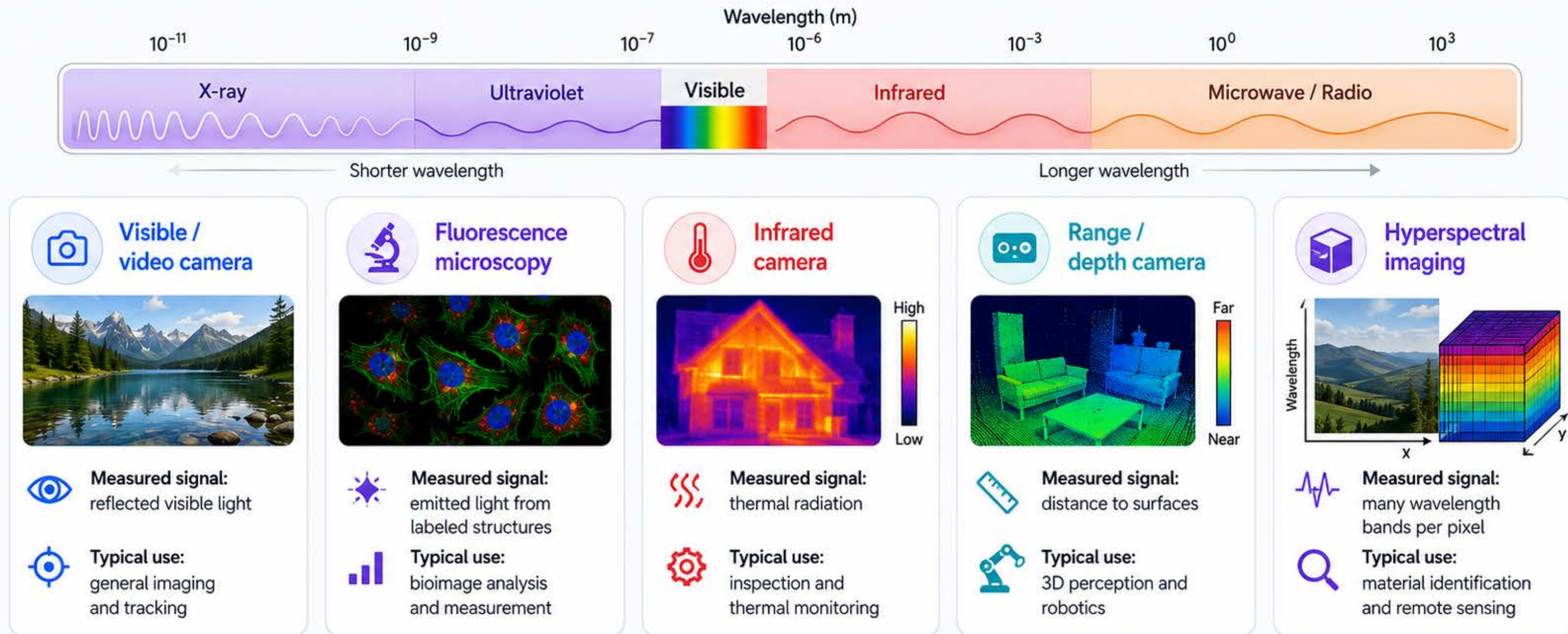
Determines which wavelengths pass.



Optics define what the sensor can see; the sensor defines what the computer can measure.

Imaging Modalities and Measured Signals

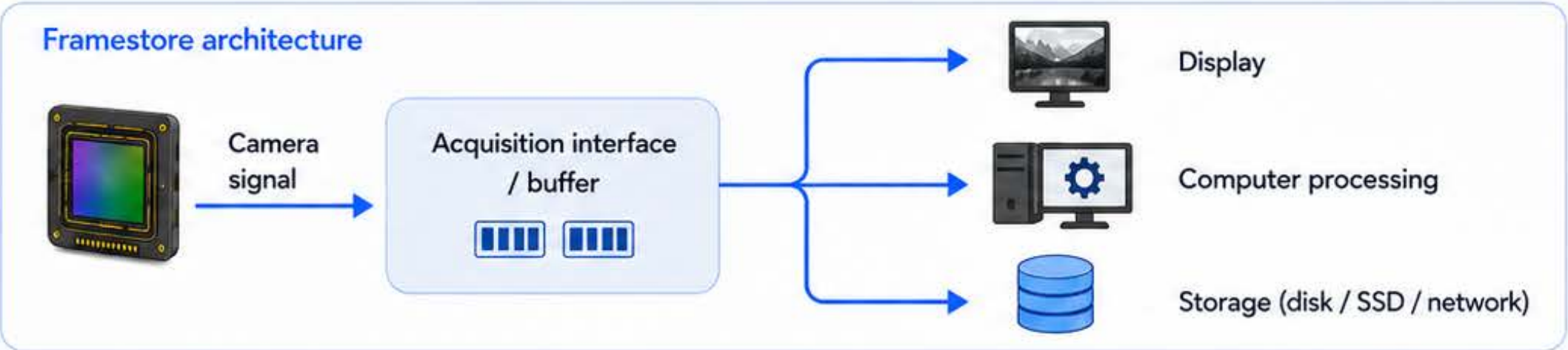
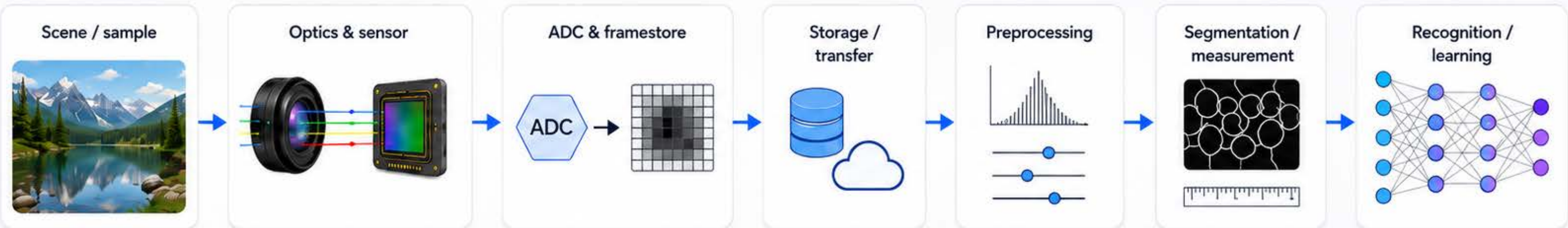
Different modalities measure different physical phenomena, so pixel values have different meanings.



A pixel value is a measurement, not just a color.
Its meaning depends on the modality, optics, and calibration.

From Capture to Computation

Acquisition connects the physical world to storage, processing, measurement, and learning.



Memory per frame
width x height x
channels x bits / 8

Video data rate
frame size x frames/s
512 x 512, 8-bit grayscale
at 25 fps ≈ 6.25 MiB/s

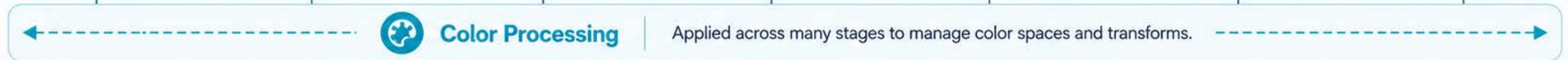
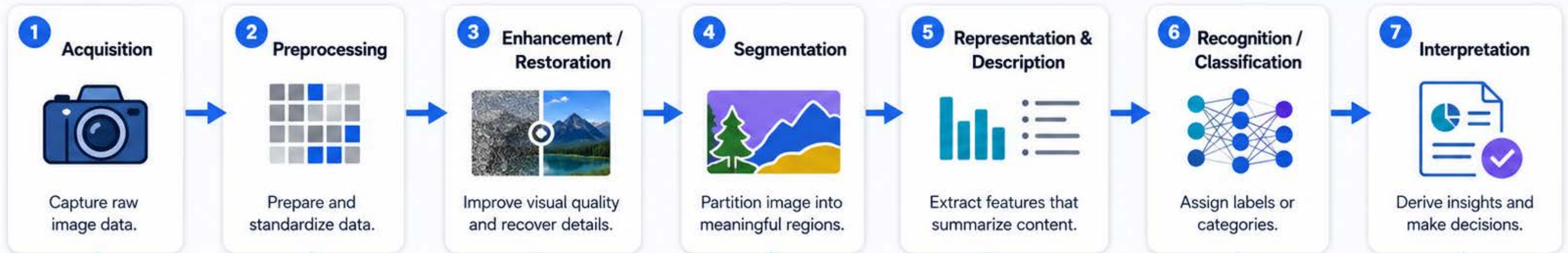
Why buffering matters
If data arrive faster than
they can be stored or
processed, frames may
be dropped.

Why quality matters
Acquisition quality shapes
enhancement, segmentation,
measurement, and learning.

A good workflow preserves both image content and the conditions under which it was acquired.

Digital Image Processing Pipeline

Turning raw image data into useful visual, quantitative, and semantic information.



Task-dependent workflow
No single pipeline fits every application.

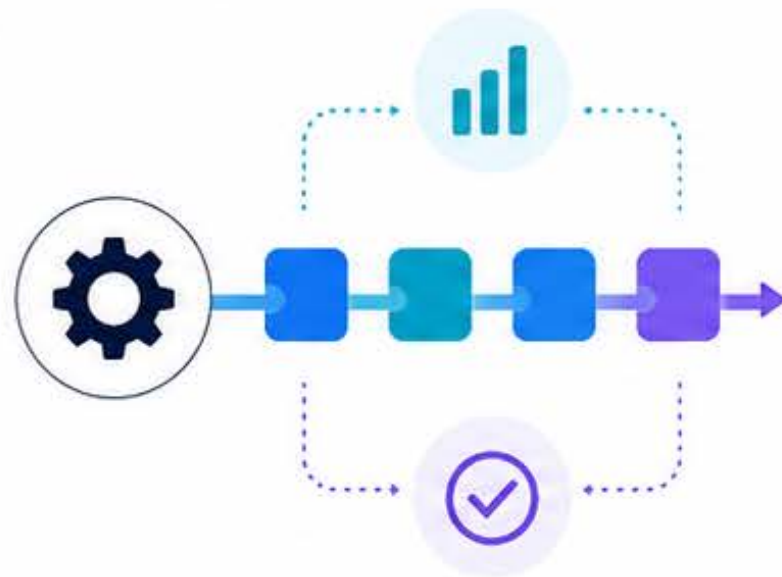
Errors propagate
Early mistakes affect later measurements and decisions.

From pixels to meaning
The pipeline gradually converts image data into conclusions.

 **A pipeline organizes image processing so each stage has a clear purpose and output.**

Why Think in Pipelines?

A pipeline links task goals, domain knowledge, and processing decisions.



A digital image processing pipeline is an **organized sequence of operations** that converts raw image data into improved images, measurements, labels, or interpretations.



Problem domain

The real-world context that defines what matters.



Imaging modality (camera, microscope, scanner)



Analysis goal (measure, detect, classify, interpret)



Acceptable errors (tolerance, uncertainty, risk)



Constraints (speed, storage, cost, safety, regulations)



Knowledge base

Prior information that constrains and informs decisions.



Expected sizes, shapes, and structures



Textures, colors, and intensity patterns



Anatomy, materials, or process characteristics



Microscope settings, calibration, metadata



Examples and annotations from earlier images



Task goals

What do we want to achieve?



Domain knowledge

What do we already know?



Processing choices

Which operations and parameters fit best?



Outputs

Improved images, measurements, labels, or interpretations.



Why pipelines help

Break complex problems into clear steps, so purpose, testing, and validation are explicit at each stage.



Task-dependent design

Photography, microscopy, and inspection have different goals—so stage priorities differ.



Knowledge guides choices

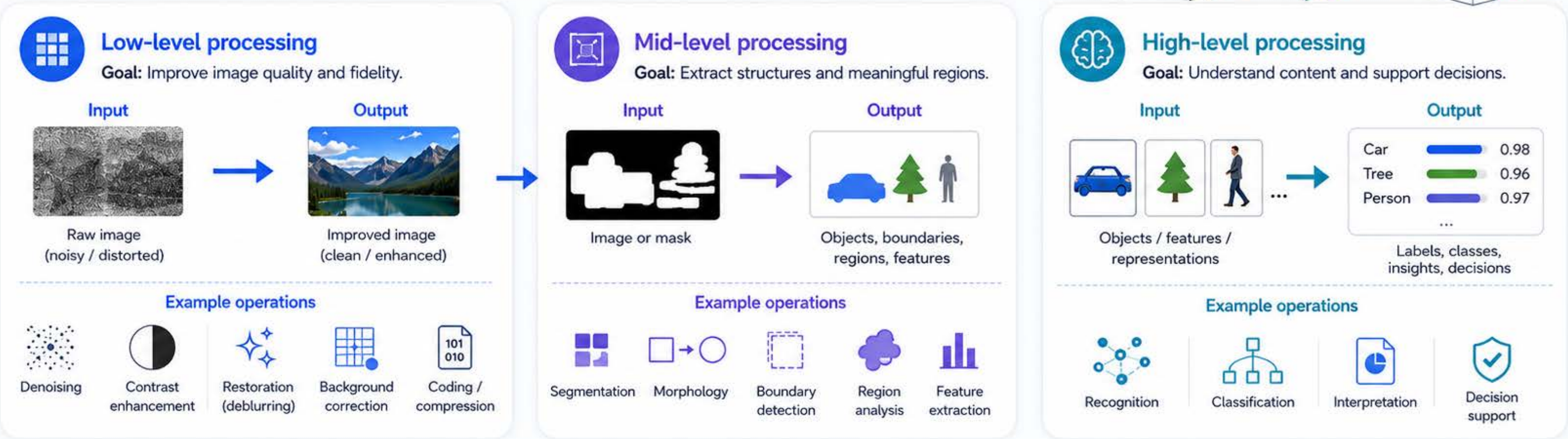
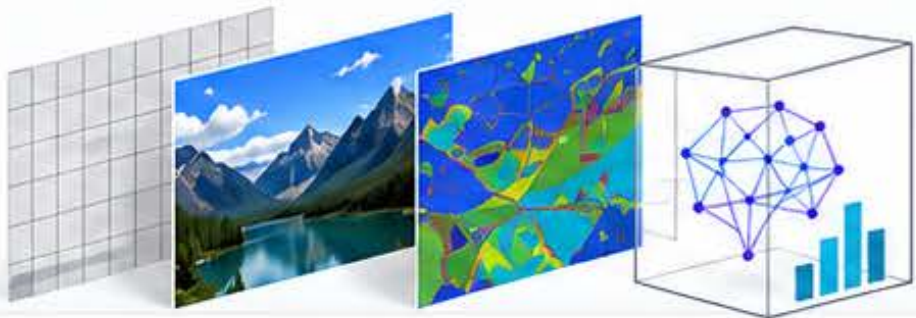
Domain knowledge can tune denoising, thresholds, segmentation, and classification for better results.



The best pipeline depends on the task, the data, and the prior knowledge available.

From Pixels to Meaning

Low-level, mid-level, and high-level processing transform image data step by step.



A pipeline becomes more semantic as it moves from corrected pixels to recognized objects and decisions.

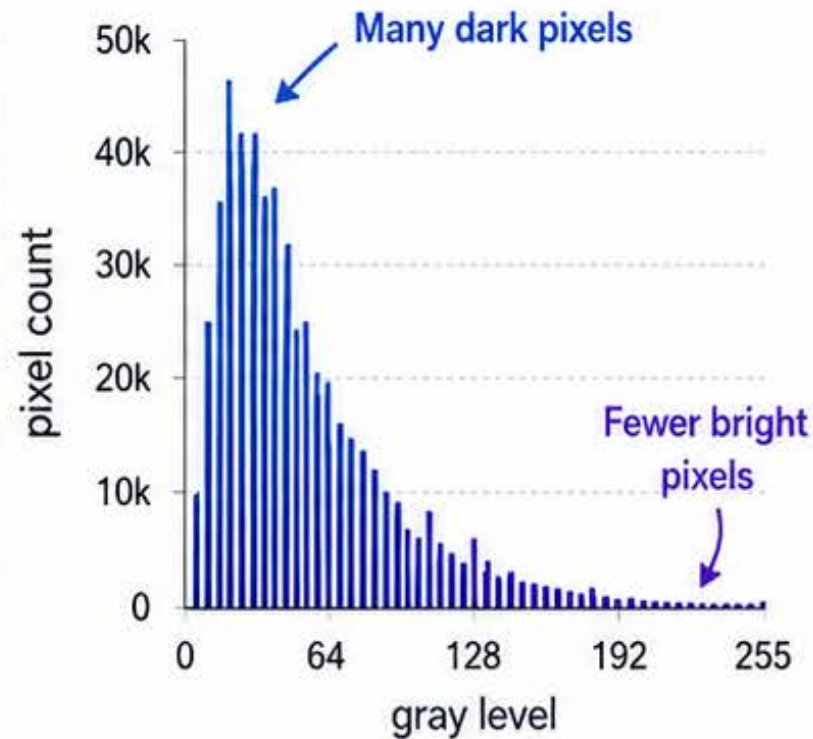
Understanding the Gray-Level Histogram

A histogram summarizes how image intensities are distributed across the available gray scale.

Example image and its histogram



Grayscale image



Gray-level histogram

Counts how many pixels fall at each intensity.



Bin

One interval on the gray-level axis.



Dynamic range

Span of available or used intensities.



Normalized histogram

Counts divided by total pixels for comparison across image sizes.

How to read it



Peak =
common intensity range.



Wide spread =
broader contrast range.



Clipping at ends =
lost detail.



Shape suggests
hypotheses, not
spatial meaning.



Important limitation

A histogram does not show where pixels are located, so it must be interpreted together with the image.



Histograms describe intensity distributions, not image geometry.

Algebraic Image Operations

Combining aligned images pixel by pixel to reduce noise, suppress background, detect change, apply masks, and build ratio images.

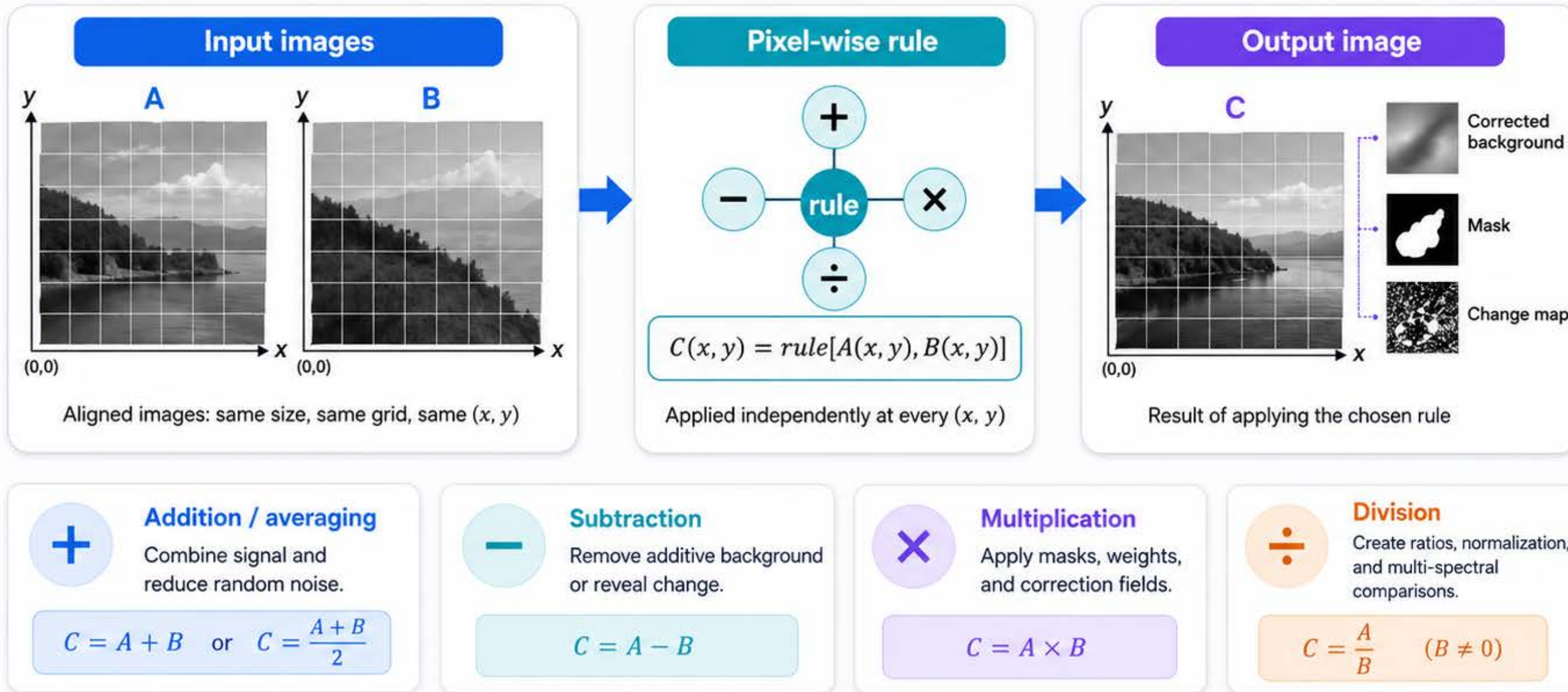


Image arithmetic is simple in form, but reliable results depend on alignment, dynamic range, noise, and calibration.

How Algebraic Operations Fit in the Processing Pipeline

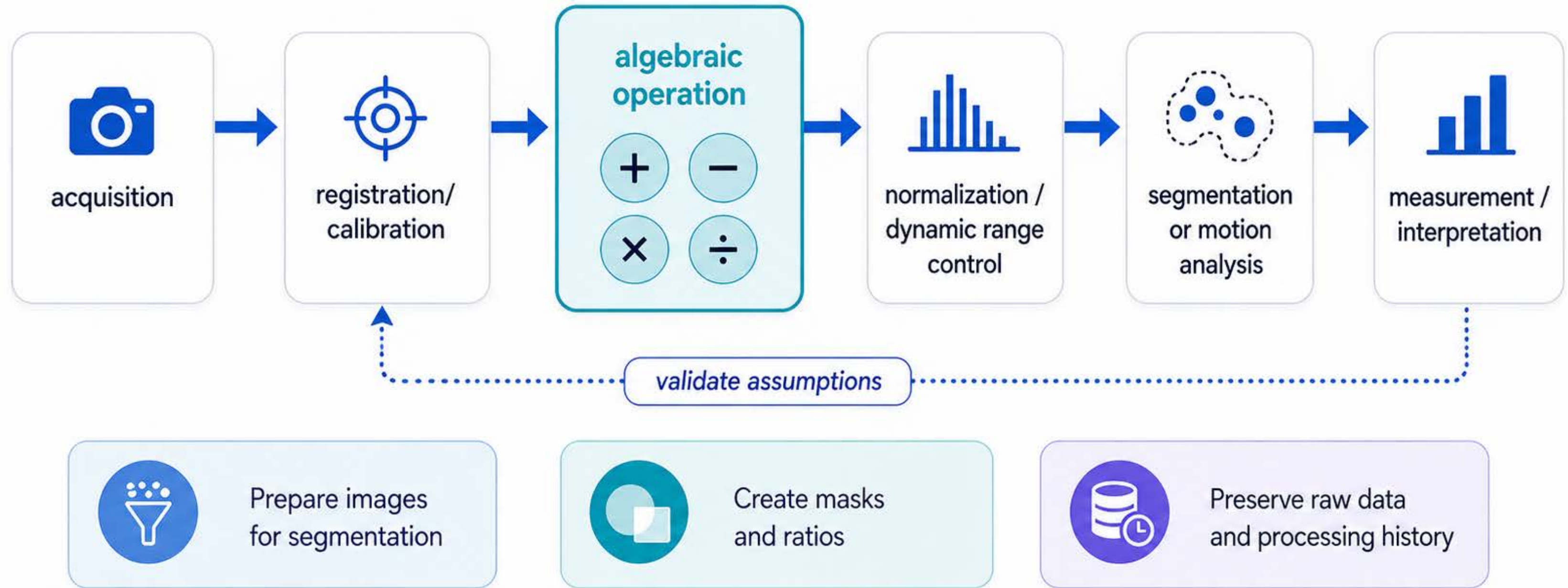


Image arithmetic is often an early transformation whose effects propagate downstream.

The Four Core Arithmetic Rules

Mathematical and operational principle



Addition

Combine corresponding pixel values.

$$C(x, y) = A(x, y) + B(x, y)$$

Example: Addition ($C = A + B$)

A (input)

20	40	60
30	80	100
10	50	90

B (input)

5	10	20
15	20	25
0	10	30



$C = A + B$ (output)

25	50	80
45	100	125
10	60	120



Compute in a suitable numeric type before clipping, rescaling, or display.



Subtraction

Remove additive background or reveal change.

$$C(x, y) = A(x, y) - B(x, y)$$



Multiplication

Apply masks, weights, and correction fields.

$$C(x, y) = A(x, y) \times B(x, y)$$



Division

Create ratios, normalization, and multi-spectral comparisons.

$$C(x, y) = \frac{A(x, y)}{B(x, y)}$$



The formula is simple; the numeric format and scaling determine whether the result is trustworthy.



Multiplication and Division: Select, Weight, Normalize

1 Binary mask multiplication

Image $\times$ mask = region of interest

Image A



Binary mask (M)



$\times$

$=$

Result C



■ 0 outside
□ 1 inside

$$C = A \times B$$



mask or weight pixels

Select which pixels contribute.
0 removes; 1 keeps.

2 Weighted mask (windowing)

Image $\times$ smooth window = tapered edges

Image A



Smooth window (W)



$\times$

$=$

Result C



0 weight 1

$$C = A \times B$$



correct multiplicative variation

Weight pixels to reduce edge discontinuities and spectral leakage before Fourier analysis.

3 Division / ratio imaging

Channel A $\div$ channel B = ratio map

Channel A
(Numerator)



Channel B
(Denominator)



$\div$

$=$

Ratio map C



High Ratio
Low Ratio

⚠ Check small or dark denominator pixels.
Use ϵ to avoid division by zero and noise blow-up.

$$C = \frac{A}{B + \epsilon}$$



compare relative channel response

Normalize out multiplicative effects (illumination, gain, thickness) to reveal relative differences.



Multiplication changes contribution; division creates relative measurements and needs denominator checks.

Key Learning Points



Pixel-wise logic

Output pixels come from corresponding aligned inputs.



Addition

Blends images and supports noise-reducing averages.



Subtraction

Removes additive background and highlights temporal change.



Multiplication

Masks, weights, and applies correction fields.



Division

Produces ratios and normalization but is denominator-sensitive.



Validity checks

Registration, dynamic range, calibration, and raw-data preservation matter.



Before applying arithmetic, ask: Are the images aligned? Are scales compatible?
Will clipping or noise change the interpretation?

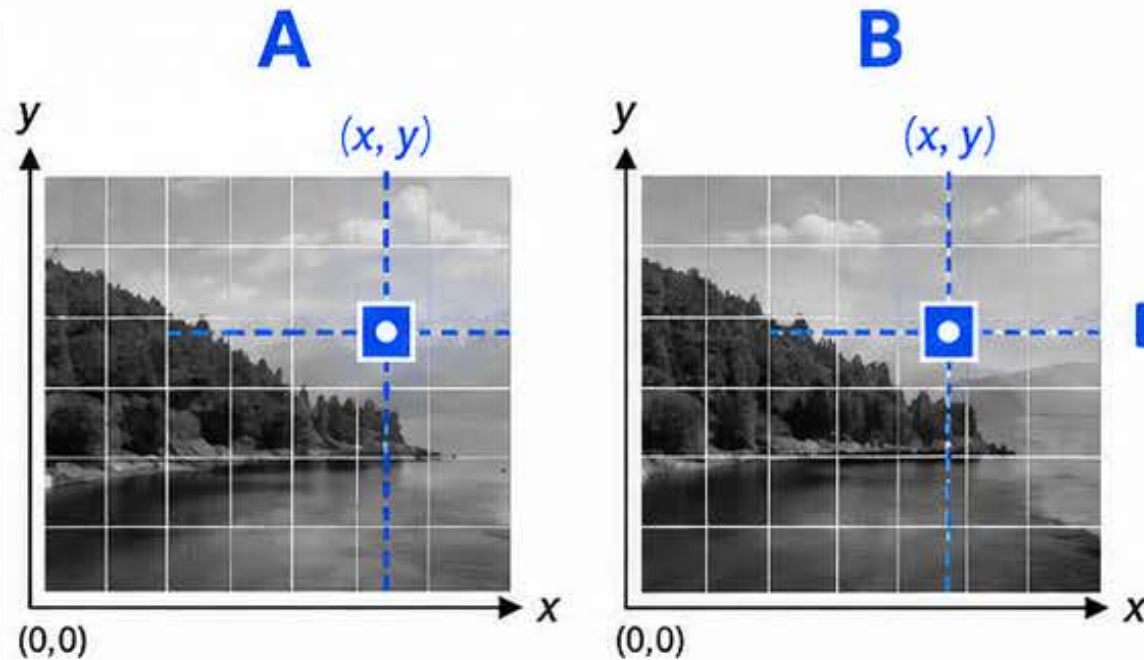


Simple arithmetic becomes reliable image processing only when its assumptions are tested.

Pixel-Wise Operations Need a Shared Grid

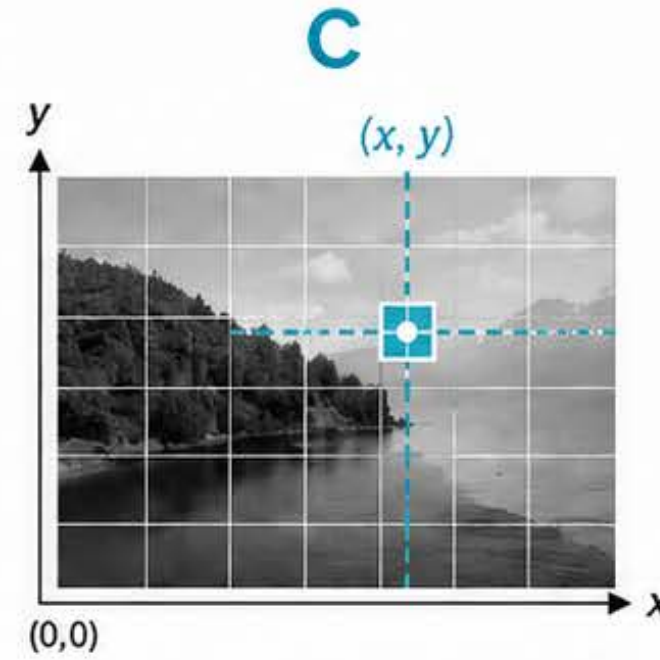
Goal: Define pixel-wise image arithmetic and distinguish it from local spatial filtering.

Aligned input images (A and B)



- ✓ Same coordinate system
- ✓ Same image size or registered geometry
- ✓ Same physical meaning at each location

Pixel-wise output (C)

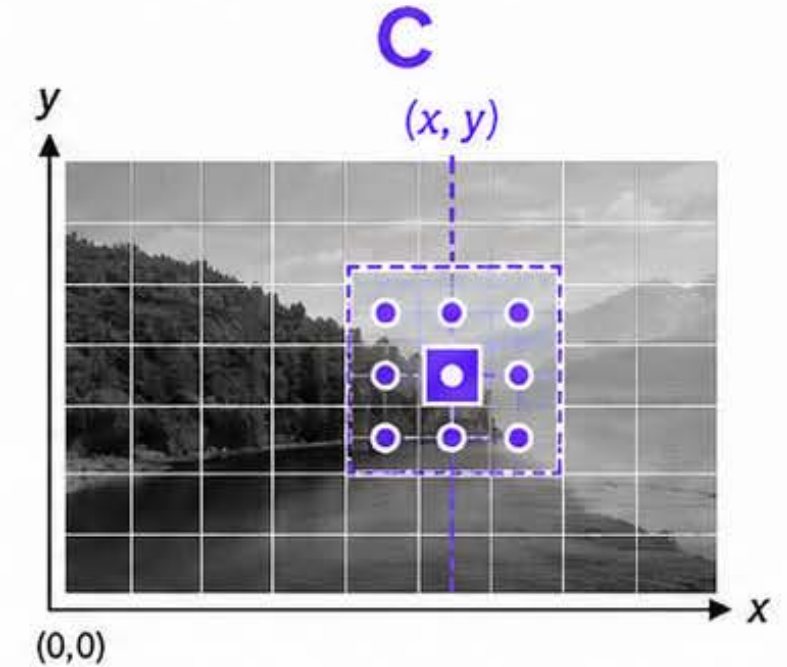


- 💡 Each output pixel is computed only from the values at the same location (x, y) .
- ✓ **No neighborhood** is used in basic algebraic arithmetic.

≠

Local filter (not the same idea)

A local filter computes the output at (x, y) using a neighborhood of input pixels.



- ✗ **Not the same idea:** uses surrounding pixels, not just the corresponding one.



Arithmetic is meaningful only when corresponding pixels represent the same scene or sample point.

Local Spatial Operations and Filtering

Local filtering uses the neighborhood around each pixel.
It brings spatial context to smoothing, noise reduction, and boundary detection.



Local filtering turns isolated pixel values into context-aware image decisions.

Thresholding: A Simple Pixel Decision

Pixels are classified by comparing image intensity with a decision value.

1. Input grayscale image $f(x, y)$

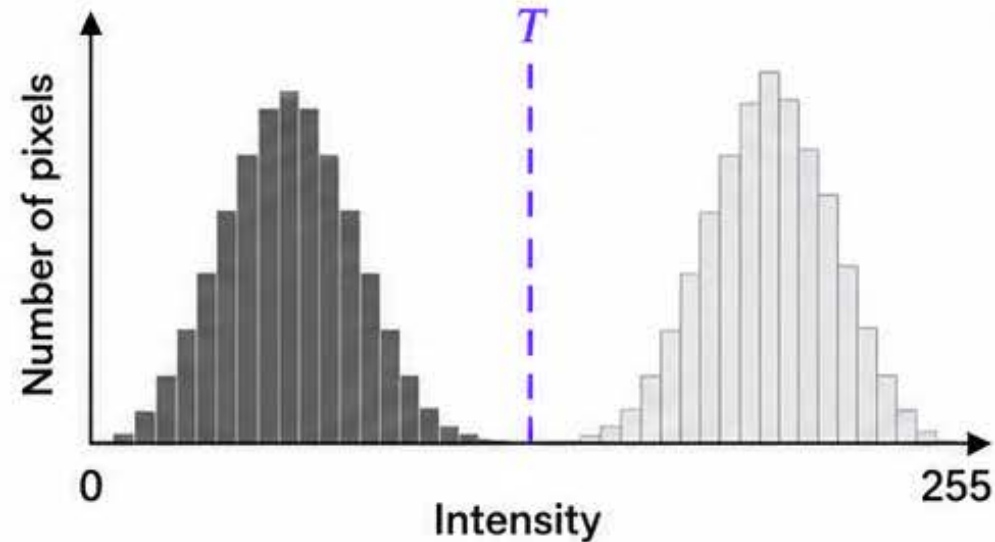
30	28	31	34	32	29
27	45	160	165	48	30
29	140	210	215	145	32
31	138	212	218	142	33
28	46	158	162	47	29
30	29	33	35	31	28



Brighter region represents the object.

2. Thresholding rule and intensity distribution

$$g(x, y) = 1 \text{ if } f(x, y) \geq T$$
$$g(x, y) = 0 \text{ if } f(x, y) < T$$



3. Output binary image $g(x, y)$

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

■ 1 = Foreground (object)
□ 0 = Background

i Global thresholding works best when object and background intensities are well separated.

 Thresholding converts gray levels into object decisions by applying a rule at each pixel.

Global vs. Adaptive Thresholding

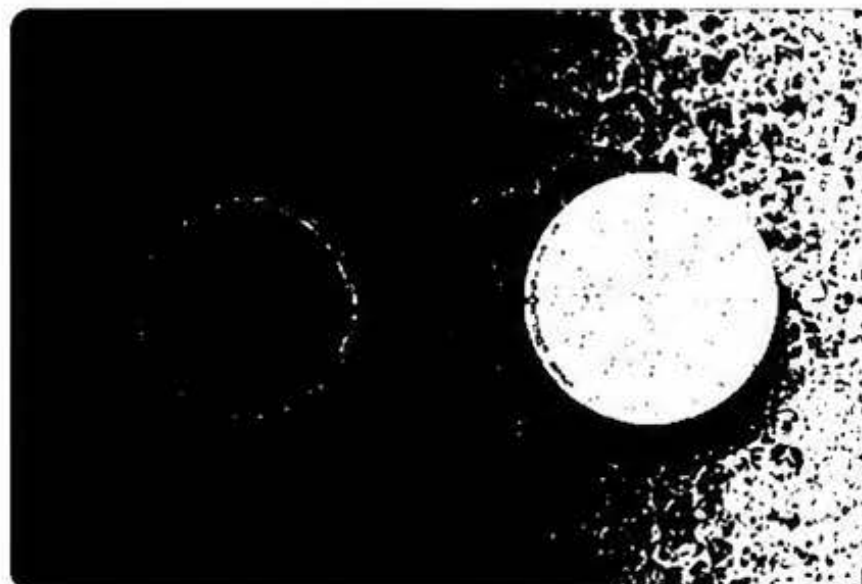
One threshold can fail when illumination changes across the image.

1 Uneven illumination



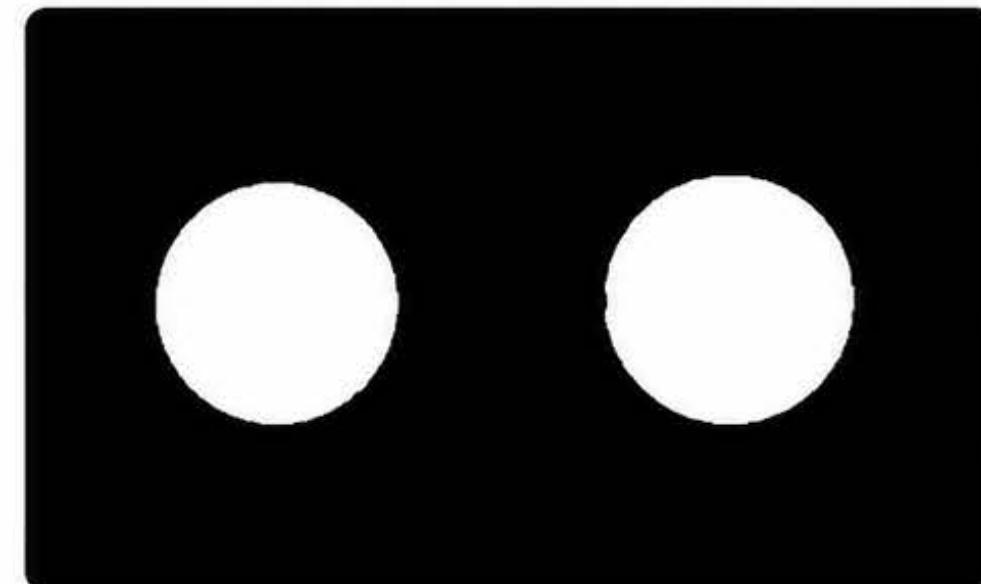
Same objects, changing background.

2 Global threshold



- Simple and fast.
- Fails when illumination varies.

3 Adaptive threshold



- Threshold varies with location $T(x,y)$.
- Sensitive to window size and noise.



Global threshold uses one T for all pixels.



Adaptive threshold uses local statistics.



Adaptive thresholding handles uneven illumination, but it must be tuned to the local image statistics.

Edge-Based Segmentation

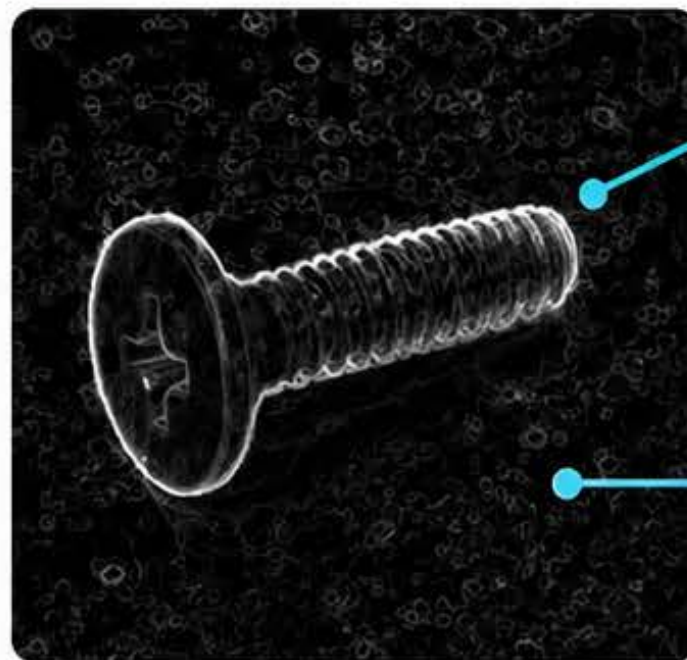
Boundaries are inferred from rapid local intensity changes.

1 Input Grayscale Image



Input image

2 Edge Strength (Gradient Magnitude)

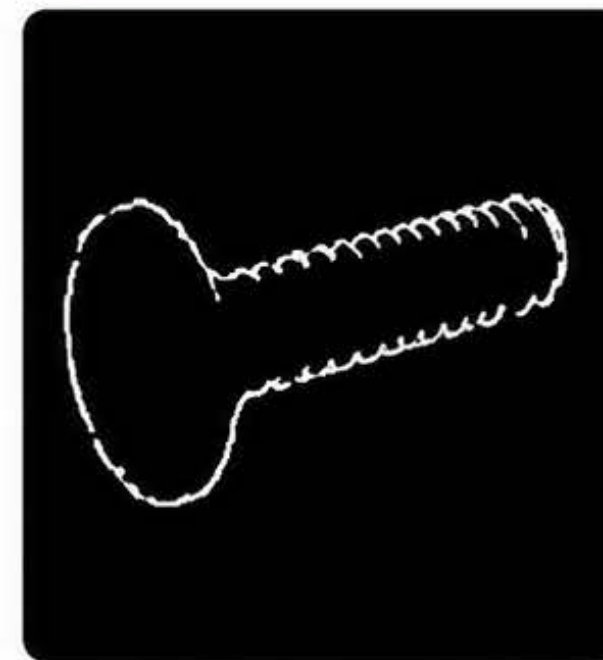


Strong
response
near
boundaries

Weak
response in
uniform
regions

Bright values indicate strong intensity changes.

3 Boundary Map



- Useful for boundary localization.
- May produce gaps, false edges, or noisy contours.



Later cleanup may use boundary tracking, morphology, or closing to form usable regions.



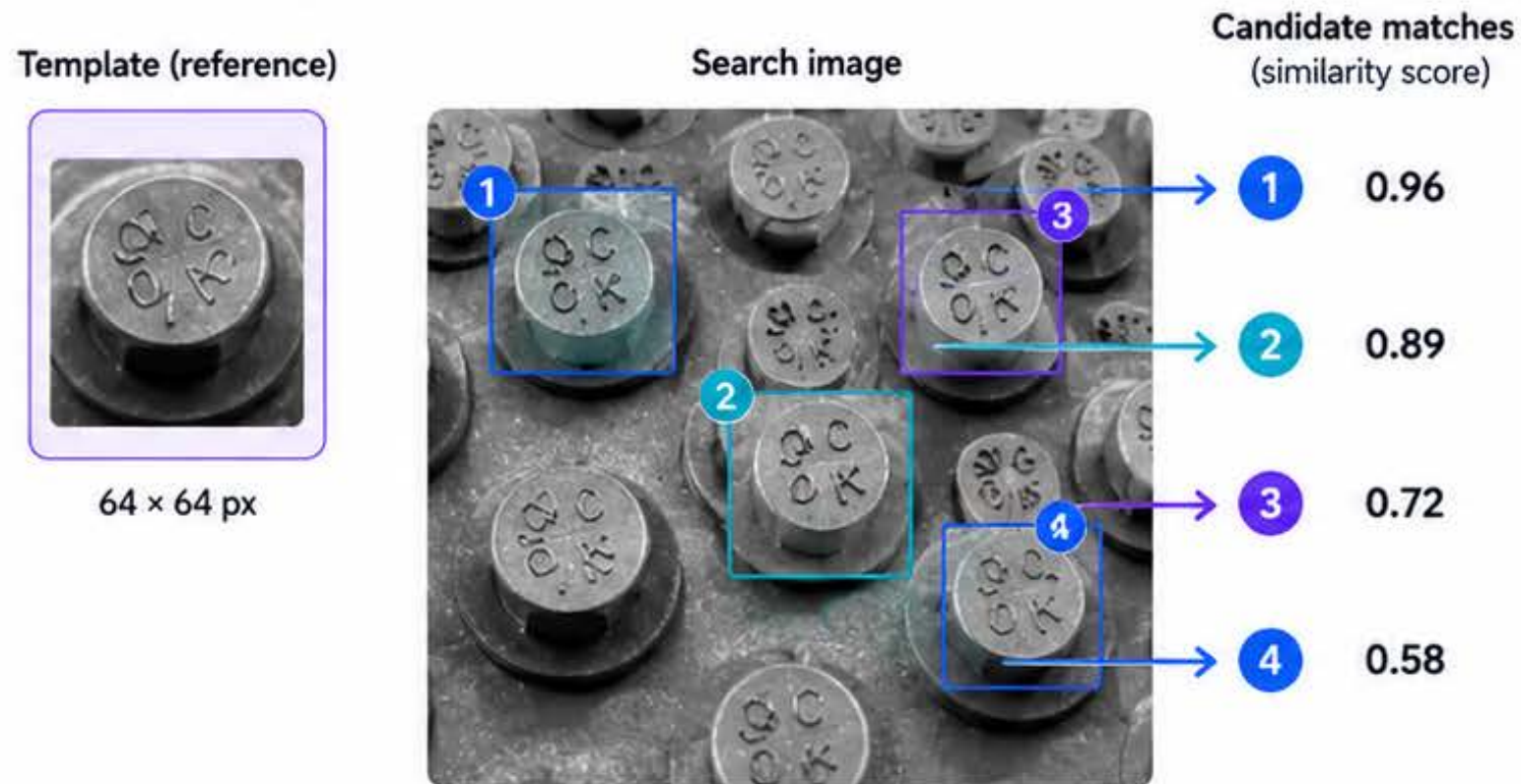
Edge evidence helps locate boundaries, but edges often need cleanup before they become usable object regions.

Template Matching and Rule-Based Recognition

Two classical ways to turn visual patterns into decisions

1. Template Matching (Similarity Search)

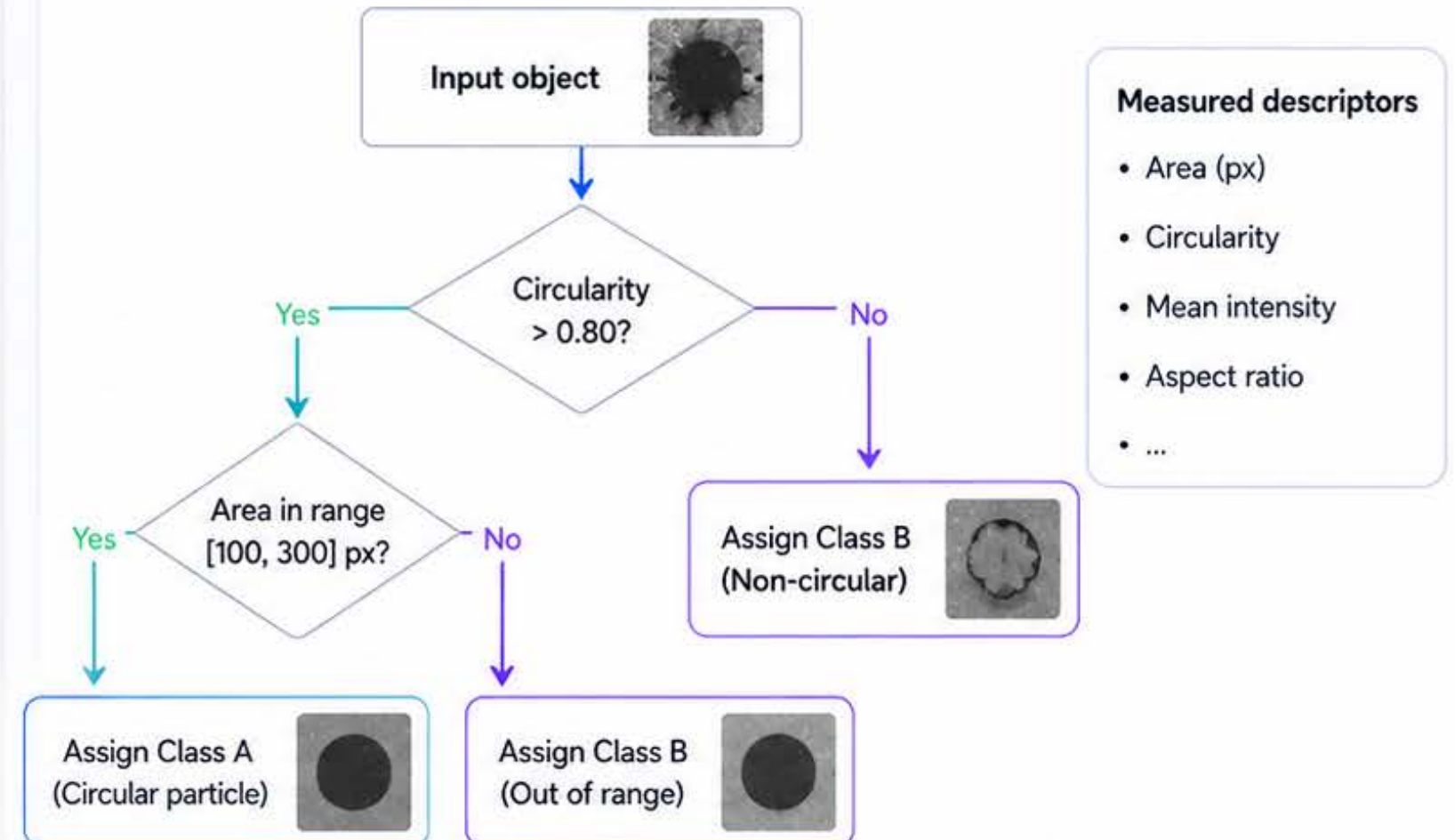
Find locations that look most like a reference template.



Higher scores indicate stronger matches.

2. Rule-Based Recognition (Decision Logic)

Use measured properties and explicit rules to assign a class.



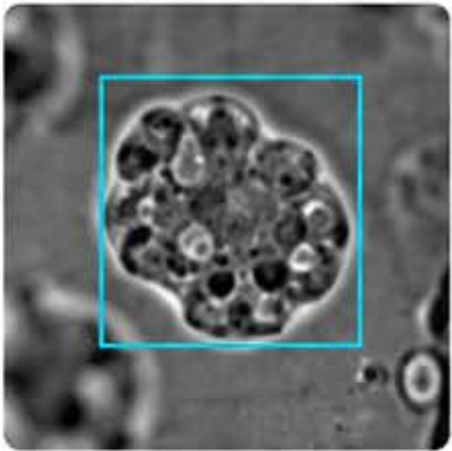
Classical recognition can rely on similarity scores, explicit rules, or both.

From Features to Decisions

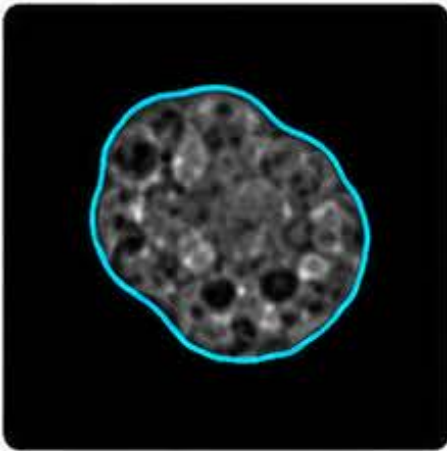
Descriptors, feature vectors, and classifier outputs

1 Object & Segmentation

Object Crop



Segmentation (Mask)



Detect and isolate the object of interest.

2 Measurements (Descriptors)

	Area	153.2 μm^2
	Perimeter	45.6 μm
	Circularity	0.73
	Mean Intensity	118.7 (a.u.)
	Texture (GLCM Contrast)	42.1

Quantify the object with descriptive features.

3 Feature Vector & Classifier Decision

Feature Vector

$x = [\text{area, perimeter, circularity, intensity, texture}]$

Classifier Output

 Predicted Class
Neutrophil

 Confidence
0.94

Assemble features into a vector and use a classifier to predict the class.



Why It Matters

Good features capture meaningful variation. Robust classifiers turn those features into reliable, confident decisions.



Recognition converts measured descriptors into evidence-based decisions.

Recognition Tasks in Image Analysis

Classification, detection, identification, and verification



CLASSIFICATION

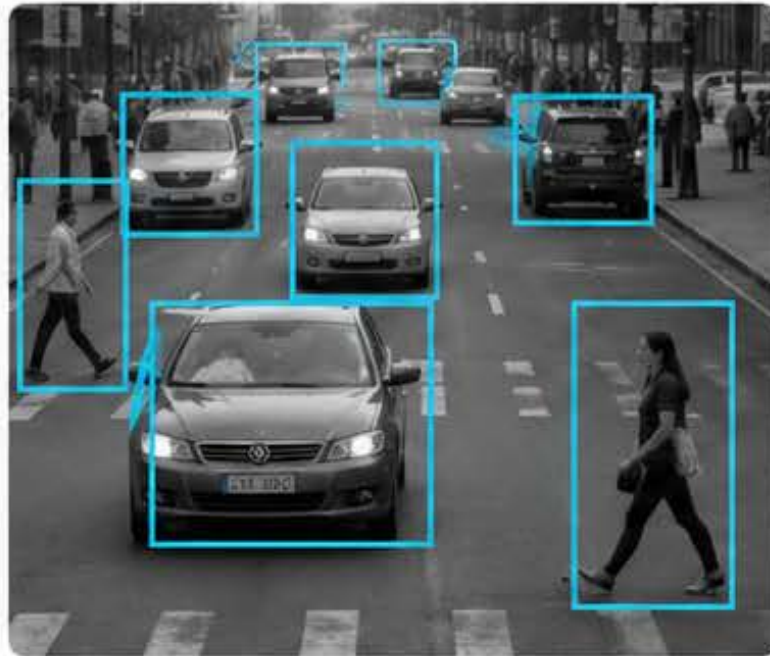


One object or scene assigned to a class.

Example: Chest X-ray → Pneumonia



DETECTION



Locate and draw bounding boxes around one or more objects.

Example: Autonomous driving → Vehicles, people



IDENTIFICATION



Determine the identity of an object or individual from image data.

Example: Fingerprint → ID: 0457A91C



VERIFICATION

Probe

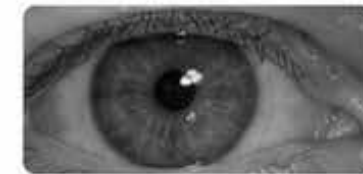


Claim



Match Score: 0.87

Result: Same



Match Score: 0.12

Result: Not Same

Determine whether two images are from the same source or individual.

Example: Iris verification → Same / Not Same



Different recognition tasks answer different questions about image content.

Machine Learning for Image Analysis

From image-derived measurements to prediction, discovery, and decision support



Why it matters

- Image data can be large, variable, and difficult to interpret with fixed rules.
- Machine learning learns patterns from measured image evidence.
- Outputs can be classes, numerical predictions, groups, or decision scores.



Core workflow



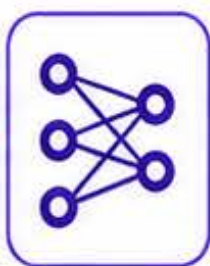
Image



Preprocess /
Segment



Measure
Features



Train or
Apply Model



Prediction /
Decision

Classical image processing produces the evidence that ML uses.



Learning focus

- Features, feature vectors, and feature tables
- Supervised and unsupervised learning tasks
- Validation, metrics, and practical risks



Machine learning adds value when reliable image measurements are transformed into actionable outputs.

Supervised Learning

Labeled examples support classification and regression

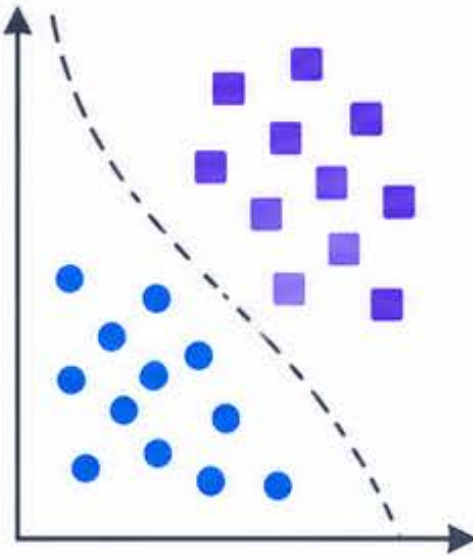


Classification

Predicts a discrete category

$$\hat{y} = C(x)$$

- normal vs abnormal
- cell type A vs B
- pass vs fail



Regression

Predicts a continuous numerical value

$$\hat{y} = f(x)$$

- response score
- lesion volume
- material thickness



Common classical methods



k-nearest neighbors

simple baseline



support vector machine

strong boundary-based classifier



decision tree

interpretable rules



random forest

robust ensemble of trees



regression models

useful for quantitative prediction



Supervised learning depends on representative labels and reliable ground truth.

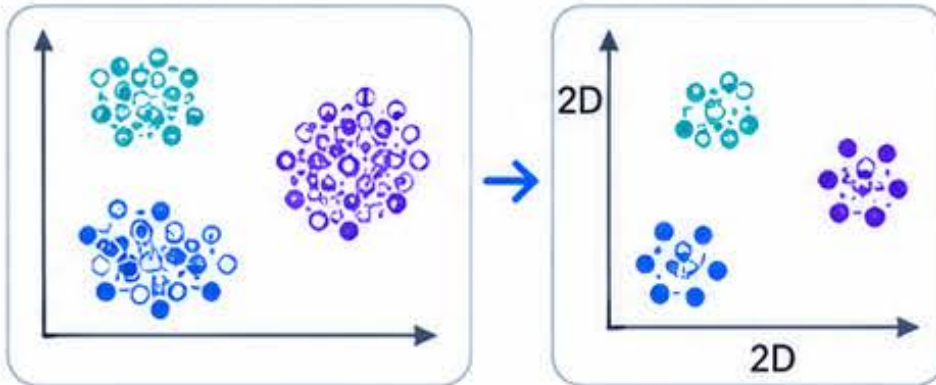
Learning with Limited or No Labels

Unsupervised, semi-supervised, and self-supervised approaches broaden the toolbox



Unsupervised learning

- Clustering groups similar samples without class labels
- Dimensionality reduction summarizes or visualizes high-dimensional data
- Useful when labels are unavailable or ambiguous

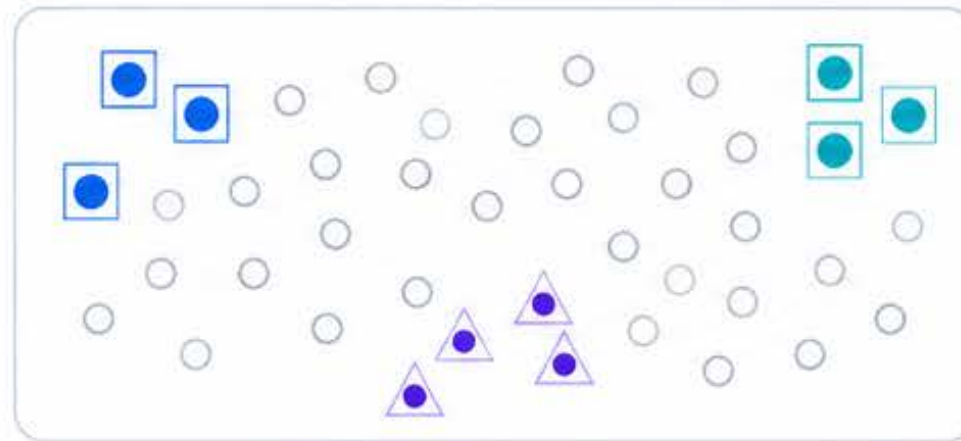


Clusters are hypotheses, not automatic truths.



Semi-supervised learning

- Uses a small labeled set plus many unlabeled samples
- Helps when annotation is expensive
- Can improve learning, but error propagation is a risk

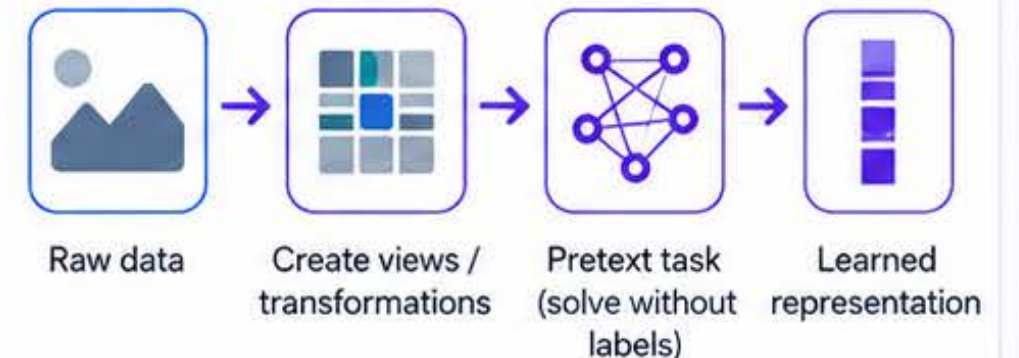


● Labeled (Class A) ● Labeled (Class B) ● Labeled (Class C)
○ Unlabeled



Self-supervised learning

- Creates supervisory signals from the data itself
- Learns representations from structure, time, or transformations
- Useful before a downstream prediction task



When manual labels are scarce, structure in the data can still guide learning.

Validation, Metrics, and Common Risks

Reliable performance requires unseen data, appropriate metrics, and careful experiment design



Data split and generalization



- Training data fit the model
- Validation data tune model choices
- Test data stay unseen until final evaluation
- Generalization means working well on new data



Confusion matrix and metrics

		Predicted	
		Positive	Negative
Actual	Positive	TP	FP
	Negative	FN	TN

- Accuracy = $(TP + TN) / \text{all}$
- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
- F1 = $2PR / (P + R)$

Accuracy can be misleading when classes are imbalanced.



Practical risks



Overfitting



Underfitting



Class imbalance



Data leakage



Domain shift



Annotation noise / poor segmentation



A trustworthy model must be evaluated on independent data that reflect the real use setting.

Summary and Learning Roadmap for Image Processing

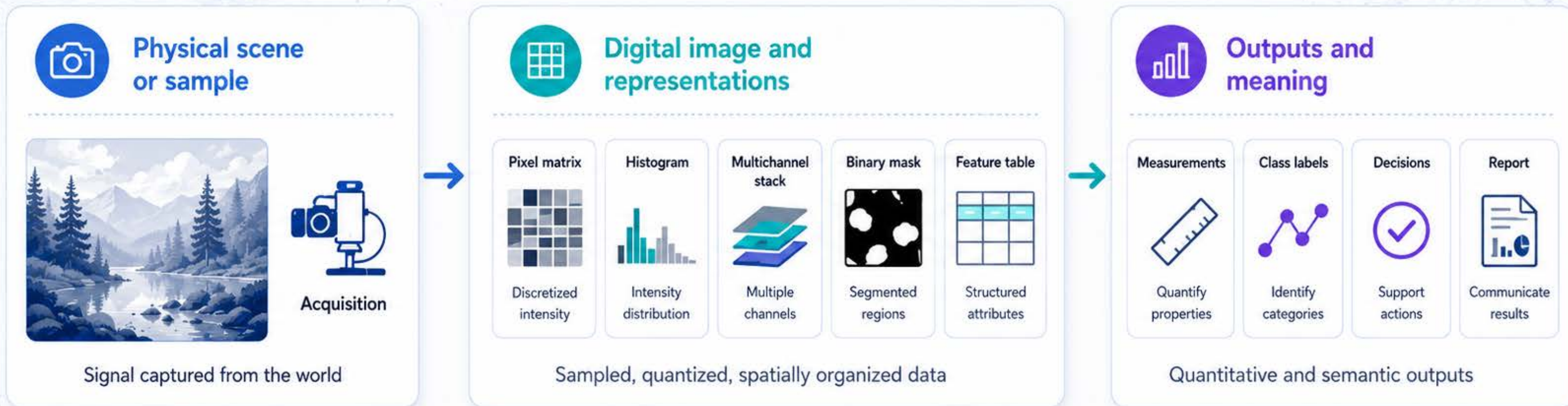
From pixels to validated insight



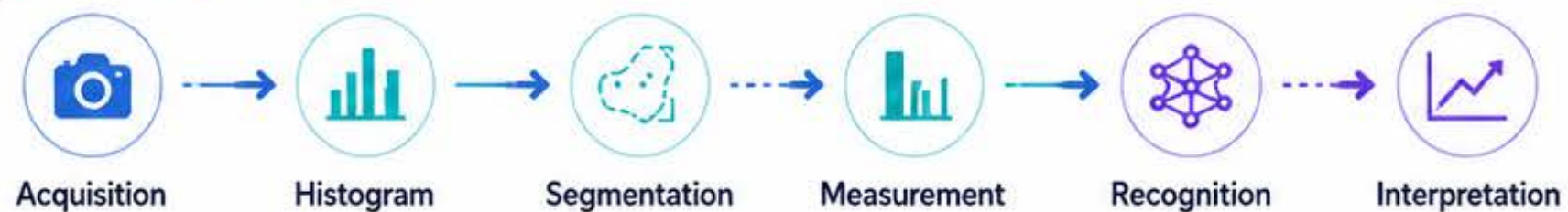
Goal: produce **evidence**, not just better-looking images.

The Complete Image Processing Journey

An image is data that moves through a connected pipeline



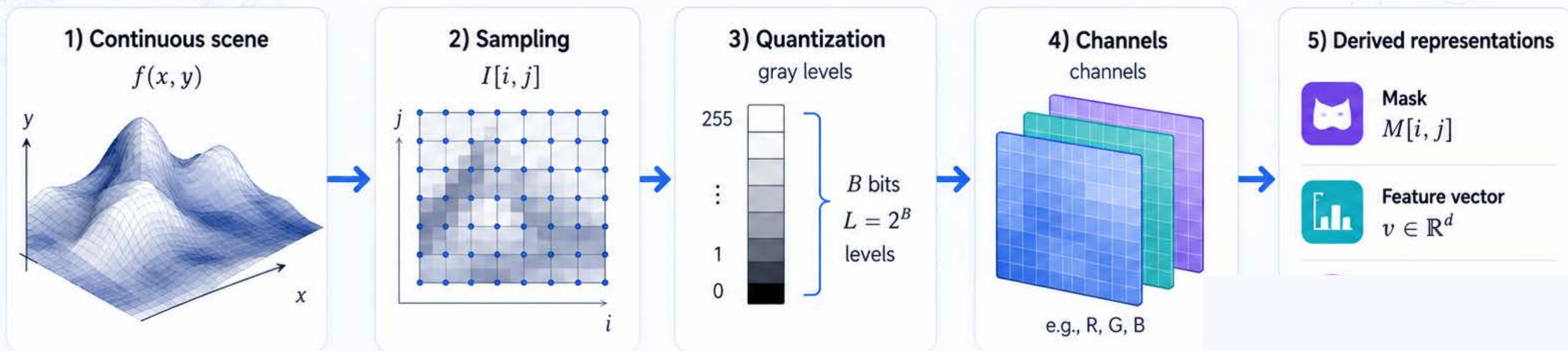
No stage is isolated



- Every stage transforms the data.
- Early choices constrain later analysis.
- Reliable interpretation depends on the whole chain.

From Continuous Signal to Digital Representation

The mathematical and operational principle behind digital images



continuous signal $\rightarrow$ sampling + quantization $\rightarrow$ finite array



raster grid
regular sample points



bit depth
finite gray levels



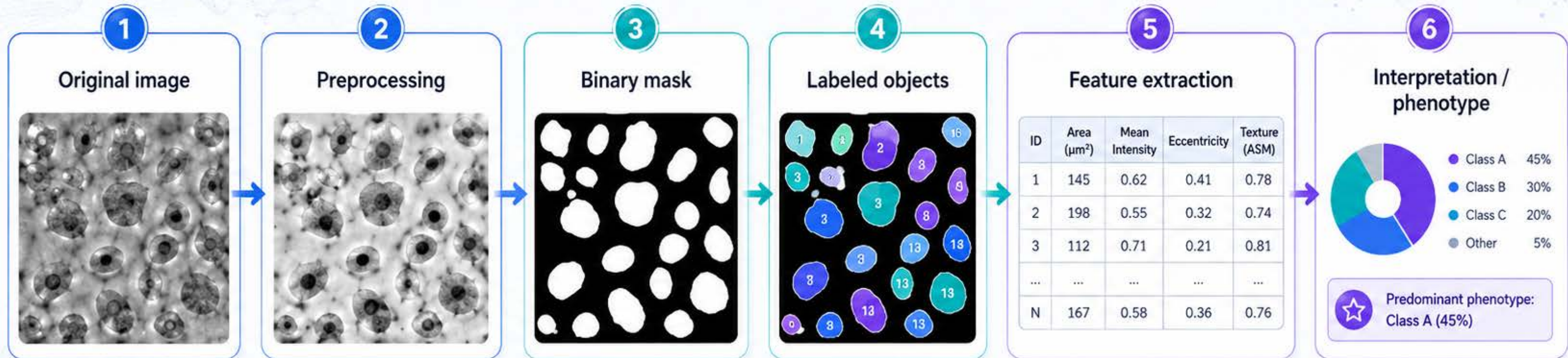
channels
multiple measurements
per pixel



metadata
context about
acquisition

From Image to Measurable Objects

A clean visual example of analysis



segmentation creates objects

convert pixels into separate measurable regions



features create evidence

area, intensity, shape, texture



interpretation needs context

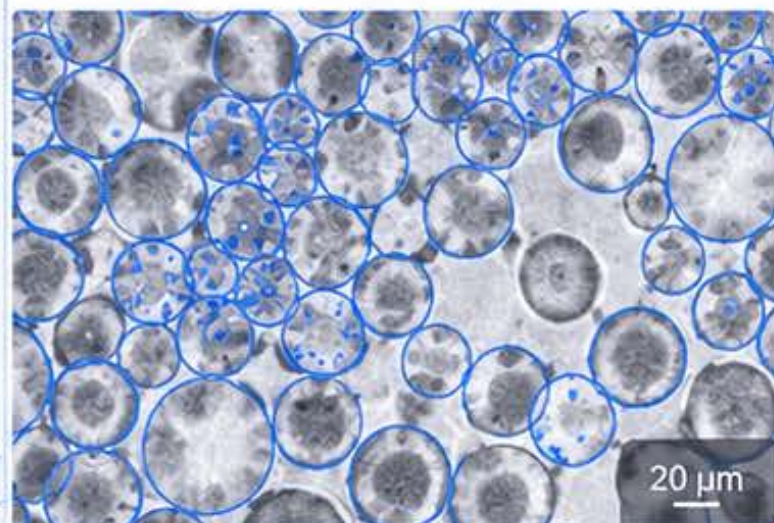
domain knowledge turns numbers into meaning

Applications Share a Common Logic

Different domains, same analytical structure



Microscopy
measurement



Outputs



Masks



Counts



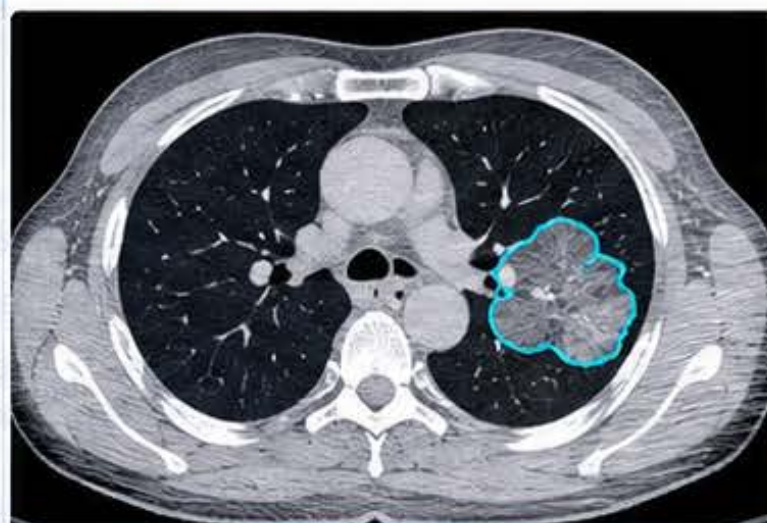
Measurements



Reports



Medical image
analysis



Outputs



Masks



Quantification



Labels



Reports



Industrial
inspection



Outputs



Defect
masks



Counts



Class
labels



Reports



Computer vision /
screening



Outputs



Detections



Tracks



Attributes



Reports



Different domains, same requirement: **validated evidence.**

